



FUEL MANAGEMENT Department
SIMHADRI

Note No :1



Ref:SMTTP/Fuel Mgmt./2024-25/LMI035/698639

Date:23/11/2024(13:28:15)

Subject:Approval for LMI on Guidelines For MGRRailway Siding Operation

Ref: SMTTP/Fuel Mgmt/2024-25/LMI 035

Subject: Approval for LMI on Guidelines For MGR/Railway Siding Operation

The LMI for 'Guidelines For MGR/Railway Siding Operation with Ref no: LMI/OGN/OPS/SYST/035 Rev 03 is hereby put up for approval of Competent Authority.

Changes incorporated in this revision (Rev: 3), have been mentioned and as per latest Operation Guidance note on Guidelines For MGR/Railway Siding Operation - OGN-OPS-SYST-035 Rev-1 Sep. 2024 and in detail in "3.0 Records of Revision".

Enclosures: -

1. 'LMI on 'Guidelines For MGR/Railway Siding Operation' with Ref no: LMI/OGN/OPS/SYST/035 Rev 03 dt Nov 2024
2. COS-ISO-00-OGN/OPS/SYST/035 Rev-1 Sep. 2024

Submitted for approval please

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It is requested to modify as per standard format of Simhadri LMI (Guidelines for Issue of LMIs as a Controlled Document) LMI/OGN/GEN/002 Rev:04 as uploaded on Simhadri EEMG LAN

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As discussed - Modified

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NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/SYST/035
REVISION NO: 3 Date: Nov 2024

GUIDELINES FOR MGR/RAILWAY SIDING OPERATION
NOVEMBER-2024

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	Guidelines for MGR/Railway Siding Operation
2	LMI No.	LMI/OGN/OPS/SYST/035 Rev 03 Nov 2024
3	Reference Document	COS-ISO-00-OGN/OPS/SYST/035 Dt . Sep-2024
4	Superseded Document	LMI OGN/OPS/SYST/035 Rev 02 May 2024
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TITLE	Guidelines for MGR/Railway Siding Operation
LMI No.	LMI/OGN/OPS/SYST/035 Rev 03

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1.0 INTRODUCTION

1.1 Scope

This Operation Guidance Note aims to provide guidelines for safe MGR / Railway Siding operations and lists out the standard operating mechanism, system requirement and safety precautions required during MGR/Railway operations

SUMMARY

Supply of Coal to Simhadri Thermal Power Plant is met through Railways System. In Rail fed Plant, Coal is transported through Indian Railways. Whether it is rail fed or pit head plant, Rail infrastructure is required to meet the coal requirement of Station. Simhadri is having two track hoppers and two Wagon Tiplers for coal unloading.

Safe and efficient handling of rake is paramount importance for uninterrupted supply of Coal. Operation of rakes requires setting of points in proper position, conforming non occupancy of track and exchanging signals before permitting a rake movement. Movement of rake/train can be divided into two major categories i.e., within station limits (yard) and between stations. There is defined methodology for movement of rakes for each type. Further the handling of rake in WT operation requires multiple movement attachment/detachment of wagons and parting of rake & which requires additional safe measures. Sole objective is to keep minimum cycle time without compromising with safety.

In general, NTPC broadly follows Indian Railway practices for operation & maintenance of rail infrastructure and the guidelines provided here are derived from it.

This document is revised keeping in view of present requirement of stations to provide guidelines for safe and reliable operation of MGR/Railway Siding.

BOBRN RAKES:

All the incoming BOBRN loaded rakes are dispatched from Duvvada station, with Railway Locomotives & crew, and admitted directly to TH for Unloading. The Loco with crew facilitate the placements in the THs. After completion of unloading the empty rake with Railway Loco and crew moves over the empty bulb to R&D yard and then to Duvvada depending on the line clearance.

Note: In case of occupancy of both the THs by BOBRN rakes then the next incoming BOBRN is stabled at R&D yard till an empty rake departs from TH.

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BOXN RAKES:

All the incoming BOXN loaded rakes are dispatched from Duvvada station, with Railway Locomotives & crew, and admitted directly to WT for unloading. The placement of the first wagon to the reach of side arm changer is done by Railway Locos, in pushing mode. The Loco is detached and taken to R&D yard by the crew. The wagon placements on the tippler table is done by the Side Arm Charger (SAC).

After completion of unloading the Railway Loco and crew are arranged and attached to the empty rake for drawn out. The empty rake moves over the empty bulb to R&D yard and then to Duvvada via Crossing Station depending on the line clear.

Note: In case of occupancy of both the WTs by BOXN rakes then the next incoming BOXN rake is stabled at R&D yard till unloading is completed.

BRIEF DESCRIPTION OF AVAILABLE SYSTEMS:

Station Alpha/ Numeric code	STDV/12328677 (Notified on 14/12/2001), Under WALTAIR-(Division) of East Coast Railway (Zone).
Serving Station	Duvvada (DVD)
No of Locomotives	Nil (All Coal rake movements are done by Railway Locos and their Crew)
No of Wagons	Nil (All Incoming coal wagons belongs to Indian Railways)
Type of wagons that station can be handled	Various Types of BOBRN and BOXN wagons of Indian Railways
Unloading Points	02 Numbers of Track Hoppers 02 Numbers of Wagon Tipplers
Scheme of working of trains	EOL (Engine On Load) for BOBRN wagons Non-EOL (Engine On Load) for BOXN wagons
Track Layout	60 Kg + 52 Kg rails on 60 Kg sleepers & FAN shaped layout, Points & Crossings.
Chargeable Distance from Duvvada	23.94 KM
NTPC Simhadri Crossing Station yard	02 lines (one for load & other for empty rakes).
NTPC Simhadri Reception & Despatch (R&D) Yard.	07 lines: All lines are electrified (OHE lines) Line No: 1, 2 for parking of load rakes + Light Engines Line No: 3 for Through movement of loaded rakes. Line No: 4 for Through movement of empty rakes. Line no: 5, 6 for parking of empty rakes & Pilot Wagons. Line No: 7 for parking of Sick wagons.

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R&D Yard to Track Hopper-1, 2 and to Wagon Tippler-1, 2	Single line movement with necessary interconnections & turnouts with points & crossings for movement of loaded rakes to either of the Wagon Tiplers & track hoppers.
Level Crossings	Five manned level crossings From Jaggayyapalem Station to NTPC In-Motion Weigh Bridge and 08 unmanned Level Crossings
OHE - Over Head Equipment	Entire Track from take-off point is available with OHE except at WT's table portions.
Wagon Tiplers	Wagon Tippler 1 & 2 along with Side Arm Charger (SAC) & related equipment available.
Engine escape lines at Wagon Tiplers	One dedicated engine passing line between the WT lines and two engine escape lines at pre-tipling and two engine escape lines at post-tipling available.
Empty bulb	Available from all THs, WTs and Engine line with OHE.
Loco movement at Wagon Tiplers	Electric Locos move over engine escape lines and diesel Locos can move over the Wagon Tippler table all the time.
Signalling system	Available with RRI system having 03 - Advance Starters, 06 - Home Signals, 03 - Calling on Signals, 43 - Main Signal, 13 - Post Shunt signal, 37 - Ground Shunt Signal, 64 - Points and Crossings, 03 - Derailing Switches, 108 - Track Circuits, 13 - Axle Counters. (11 SSDAC & Universal – 2 UAC)
Block Instruments	Token less Co-operative Block Instruments available at Crossing Station and R&D Cabin for Route clear Communication

2.0 Superseded Document

LMI/OGN/OPN/SYST/035, Rev No 02, May 2024, Reviewed on Nov. 2024 in line with latest revision of COS-ISO-00-OGN/OPS/SYST/035 D t . Sep-2024 Guidelines for MGR/Railway siding operation.

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3.0 RECORDS OF REVISION:

SI No	Clause No.	Sub-Heading	Rev: 1, May 2022	Rev: 2, May 2024	Rev: 3, Nov 2024
1.	2.0	Reference Documents	Not Mentioned in LMI	"COS-ISO-00-OGN/SYST/035 Rev. No.: 1, Dt. 12-Oct-2018"	
2.	3.0	Superseded Documents	Not Mentioned in LMI	"LMI/OGN/OPS/SYST/035 Rev No.: 1 Dt. May 2022"	
3.	4.4	Wagon Top Sampling	Part of LMI	Removed from LMI as it is not a part of TC Document COS-ISO-00-OGN /SYST/035 Rev. No.: 1, Dt. 12-Oct-2018. Prepared SOP and attached as Annexure of LMI "SAFE WORK PRACTICES IN ELECTRIFIED TRACK SECTION." as per TC Document COS-ISO-00-OGN/OPS/SYST/058 Rev. No.: 0 Nov. 2023 "SAFE WORK PRACTICES IN ELECTRIFIED TRACK SECTION."	
4.	4.5	Disaster Management Plan	Part of LMI	Removed from LMI as it is not a part of TC Document COS-ISO-00-OGN /SYST/035 Rev. No.: 1, Dt. 12-Oct-2018. Included as Annexure: 1 of this LMI.	
5.	5.4.2	Empty rake receiving at R&D Yard	Not Mentioned in LMI	New Introduced	
6.	7.0	Shunting staff composition	Not Mentioned in LMI	New Introduced as per TC Document COS-ISO-00-OGN/SYST/035 Rev. No.: 1, Dt. 12-Oct-2018.	

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7.	8.0	In case of accidents and derailments	Not Mentioned in LMI	New Introduced as per TC Document COS-ISO-00-OGN/SYST/035 Rev. No.: 1, Dt. 12-Oct-2018.	
8.		Syntax and Numbering System changed		Changed as per LMI/OIN/GEN/003 Rev:04 "Guidelines for Issue of LMIs as a Controlled Document " and LMI Audit	
9.		Table of contents, and other format, Syntax and Numbering System changed			Table of contents and format updated and revised as per latest LMI/OGN/OPS/SYST/035 Rev 01 Dt Sep 2024

LMI OGN/OPS/SYST/035 Rev 02 May 2024 revised as per latest revision of COS-ISO-00-OGN/OPS/SYST/035 D t . Sep-2024 Guidelines for MGR/Railway siding operation.

4.0 MANPOWER COMPOSITION, RESPONSIBILITIES AND ELIGIBILITY

4.1 Composition:

- 4.1.1. The Main Traffic Control Room** – In Simhadri MGR/FT system, control room located near TH is designated as centralized/main traffic control room, which will oversee and monitor all movement related activities of the MGR/FT system of the plant. This C/R will be manned by Traffic Controller/ Shift In-charge along with other personnel as per S&T C/R.
- 4.1.2. S&T Control Room** – In Simhadri S&T Control rooms is manned by ASM/SM and points man / shunt-man. Operation of panel for movement of trains will be carried out by: Ex-Railway ASM/SM/Shift employee. Number of points man/shunt-man for each control room is deployed as per site requirement and defined in LMI.
- 4.1.3. Train** – a train crew should consist of at least one Loco Driver and LD/ALD. Further in case of shortage of second LD/ALD, one points man/shunt-man must accompany the LD. The composition of train crew needs to be defined in LMI. Additional points man/shunt-man may also be deployed by station if required. It is not applicable for Simhadri.

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4.1.4. LC Gate – In Simhadri, LC Gate nos. 1 to 5 which is situated outside of plant premises is manned by gateman round the clock. LC gates no's: 06 to 13 situated inside the plant are unmanned due to speed restrictions.

4.1.5. Also, in Simhadri, to help in maintaining various records and housekeeping activities additional contractual staff are deployed as per system requirement.

4.2. Responsibilities

4.2.1. Traffic Controller/ Shift In-charge (Ex Railway /NTPC Employee) – will coordinate and oversee the movement of rakes among stations/railway siding to optimize the coal off take, minimize rake cycle time and demurrage (if applicable) in safe manner and maintaining the records. He will coordinate with other control rooms of MGR/FT, CHP Dept., Indian Railway's ASM/SM.

List of records & formats to be maintained at control room is given in **Annexure-I** & list of checklist is given at **Annexure-II**.

4.2.2. SM/ASM – will be responsible for managing the movement of rail traffic of the yard, under the guidance of the traffic controller, in the safest manner possible and keeping the records of each movement. He will manage the movement of rail traffic within station limits and take clearance from adjacent station for the movement of traffic between the station (block section). He will be responsible for correct settings of points and operating the signals. As far as possible he should permit movements with proper signals and in the event of signal failure, he will issue authority to proceed. He will also be responsible for issuing caution order to LD/ALD, keep proper communication with ground staff (LD/ALD/Point's man/ Shunt-man) & SM/ASM of the next station and as required for the safe movement of rail traffic.

SM/ASM will also coordinate with maintenance dept for judiciously accepting and issuing caution/block request as well as cancellation of the same related to daily maintenance work & Trolley movement.

He will also be responsible for managing required movements, shunting of wagons, identification of bulge wagons, formation of rake & handing over rake.

4.2.3. Loco Driver/Pilot (LD/LP) – will be responsible for driving the Loco/train safely on the instruction from SM/ASM & follow the enroute signals/caution/authority-to-proceed for the purpose of movement. On driving the train/loco in the block section he will be responsible for communicating with SM/ASM for departure/arrival from/to the yard and updating/ recording enroute status of Loco/Track/Signal/Gateman/patrolling staff etc. In route he will also exchange signals with LC gateman, patrolling staff, etc. and follow the caution order and other instruction for the route. During shunting in non-signaled zone, he will also

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verify the correct setting of points and communicate with ground staff for the purpose. He will also monitor the condition of Track/Loco and in general the route condition and provide feedback to the control room as required. This is not applicable for NTPC-Simhadri.

- 4.2.4. Assistant Loco Driver/Pilot (ALD/ALP)** – will assist LD in train movement, record route parameter (Loco/Track/Signal/Gateman/patrolling staff), repeat signals. He may also do the shunting activity. This is not applicable for NTPC-Simhadri.
- 4.2.5. Points man / Shunt-man** – will assist the LD in movement/shunting operation. He will also be responsible for proper & safe attachment/detachment of Loco/Wagon as per instructions SM/ASM and in communication with LD. He will set & lock the hand points in the correct position required for the movement.
- 4.2.6. Gateman** – has the responsibility of manning the Level Crossing Gate and closing it before movement of Rake/Loco/Trolley & keeping the records of movement etc. Gateman will exchange signal with LD/ALD and visually inspect the passing rake for any abnormality and inform to control room, if any observed.
- 4.2.7.** Depending on the requirement, site may require additional personal and assign/modify roles.

4.3 Eligibility

- 4.3.1 SM/ASM/LD / Shunt-man** – In Simhadri, Deployed Persons are Ex. Railway employee retired from relevant post with sufficient experience. And credentials have been verified regarding the expertise through interview/demonstration.
- 4.3.2 Points man** - In Simhadri, Deployed Persons are Ex. Railway employee retired from relevant post with sufficient experience. And credentials have been verified regarding the expertise through interview/demonstration.
- 4.3.3 Traffic Controller** - In Simhadri, Deployed Persons are Ex. Railway employee retired from relevant post with sufficient experience. And credentials have been verified regarding the expertise through interview/demonstration.

5.0 SYSTEM REQUIREMENT

5.1 Station working rule:

The station working rules (SWR) / working instruction have been prepared by station and approved from Head of the station, containing each expected movement and setting of points & signals for all such movement for all stations (yards). It should also contain approach in case of interruption of line for long duration. If a project is having more than one control room, then it should identify the main control room for

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deciding priority of movements (coordination). This SWR of each control room needs to be displayed and soft copy to be readily available to all MGR staffs.

SWR should contain instruction for following activities pertaining to the station –

- Brief introduction of layout of MGR system, Control Rooms & duties of each control room
- All expected movements with settings of points and signals
- Movement in case of enroute failure of loco/train
- Movement in case of communication failure.
- Unloading / rake placement
- Sampling / uncovering tarpaulin cover.
- Shunting movement.
- Actions in case of Derailment/Accident/other eventualities

OHE ISOLATION & NORMALIZATION: -

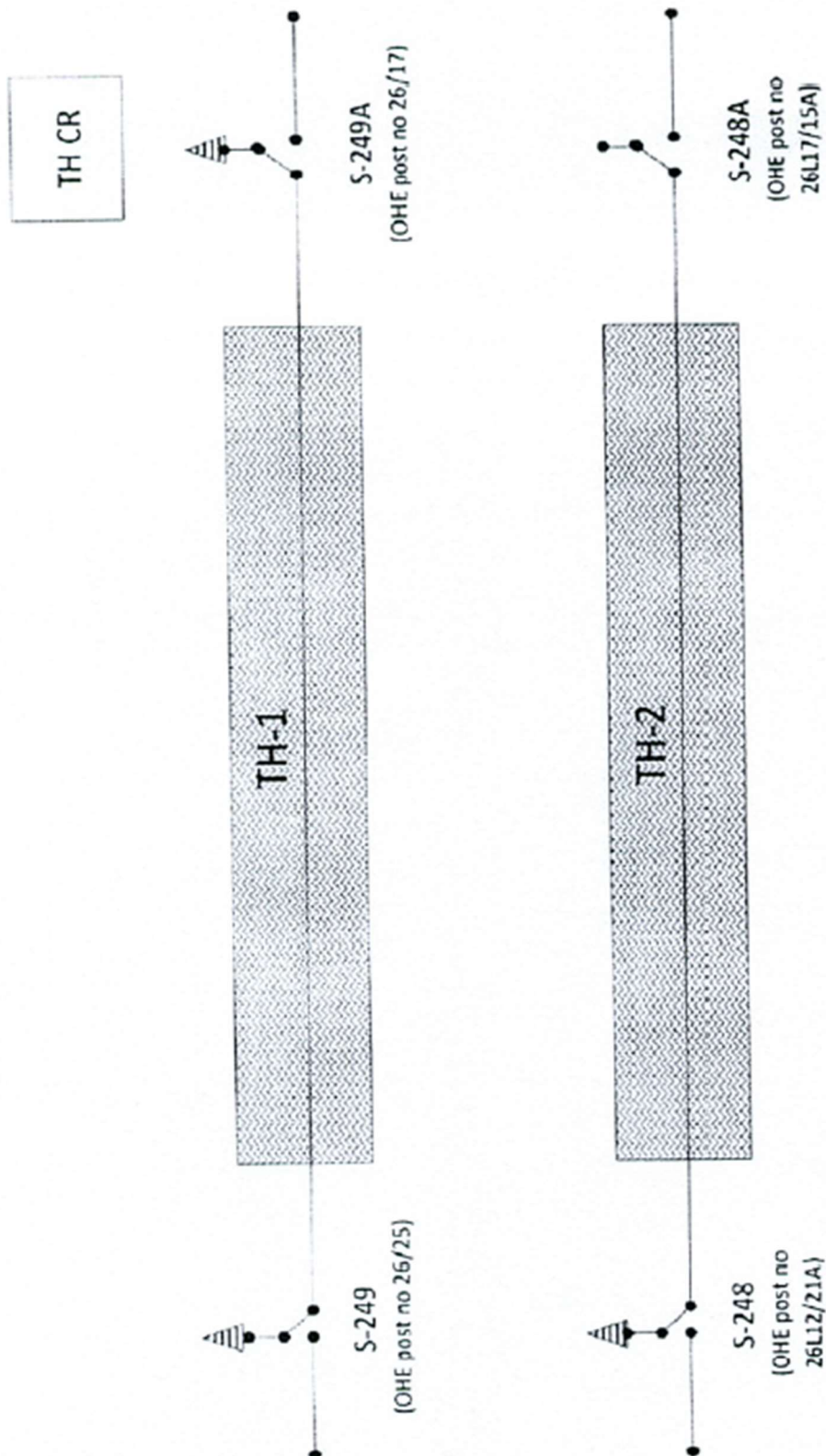
Track Hopper # 1 Isolation: - First open isolator # SS 249 (Incoming) and then open isolator # SS 249A (Outgoing)

Track Hopper # 1 Normalization: - First close isolator # SS 249A (Outgoing) and then close isolator # SS 249 (Incoming)

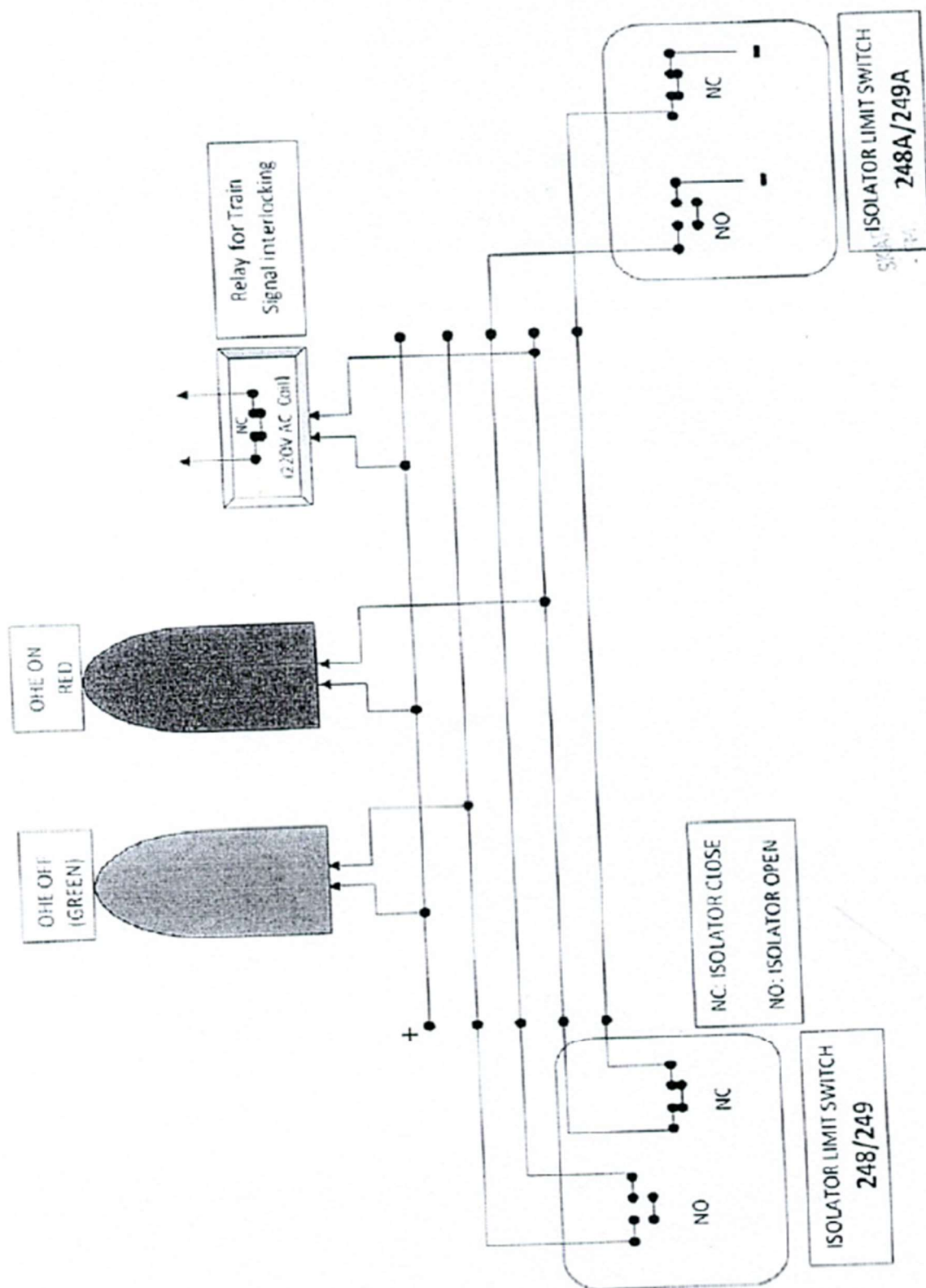
Track Hopper # 2 Isolation: -- First open isolator # SS 248 (Incoming) and then open isolator # SS 248A (Outgoing)

Track Hopper # 2 Normalization: - First close isolator # SS 248A (Outgoing) and then close isolator # SS 248 (Incoming)

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5.2 Signalling System

Signalling system ensures safe and efficient movement of rail vehicles. Simhadri station will endeavor to keep the signalling system available and working properly, if it is not available then steps will be taken to make it available.

If at any station signalling system is not available then all movement to be done manually with hand flags, SM/ASM needs to give clear instruction/shunting memo, in written, to points man /shunt-man for setting of points/movement. Points man/shunt-man must ensure that points and crossing are set and locked before any movement. LD/Points man/shunt-man must be aware of signs for using hand signal/VHF and training may be given if required.

5.3 Communication System

Communication systems are essential for safe movement of rakes. Communication system for MGR/Siding Operation purpose can be broadly divided into two categories –

- Communication between Stations & LC Gates – dedicated communication needs to be available between control rooms (stations) and between control rooms & LC.
- Communication between SM/ASM and ground staff/Loco Crew/ Trolley-man – VHF based communication devices are used for such communication. Other options such as SIM (GSM) based and License Free Walkie-Talkie are also available which can also be explored if VHF based communication is not available however, priority to be given to VHF communication to avoid the dependence on third party for network.

VHF/UHF Base stations to be provided at all control rooms & Locos. Walkie-Talkies to be used by Points man /shunt-man and maintenance personals for communication with control room. As the use of VHF equipment requires specific etiquette, training to be provided to users for the same.

Do's & Don'ts are provided at **Annexure-VIII**.

Availability and sufficiency of communication system at station to be accessed by MGR Operation In-charge and if there are any deficiency suitable measures to be taken.

To avoid the dependency on third party for network, dedicated communication for point (a) above & VHF/UHF for point (b) above to be preferred.

Important Contact Nos like main control room, hospital, first-aid, fire brigade, MGR

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operation In- charge, Local Police station to be displayed at each station.

Communication facilities available at Simhadri:

Location	Manned by	VHF	Walkie Talkies	Mobile	Rly Auto Phone	NTPC Intercom	Magnet o Phone
Crossing Station	Retired Rly staff	✓	✓	✓	x	✓	✓
R&D Cabin	Retired Rly staff	✓	✓	✓	✓	✓	✓
Track Hopper Control Room	NTPC Executive/ DT & Retired Rly staff	✓	✓	✓	✓	✓	✓
Wagon Tippler Control Room	Retired Rly staff	✓	✓	✓	✓	✓	✓
Level Crossing Gates	Retired Rly staff	✓	✓	✓	x	x	✓

Note 1 – Railway Control Phone available at Crossing Station, R&D Control Room and TH Control Room.

Note 2 – Traction Power Control (TPC) Phone available at R&D Control Room and TH Control Room. R&D Control Room and TH Control Room.

5.4 Rake Monitoring System

Rake monitoring to monitor the movement of rake/Locodata like speed, position, timings etc. are carried out to through constant touch with Controllers Railways and also through FOIS.

5.5 Movement Formats

Control rooms are maintaining records & format as mentioned at **Annexure-I**.

5.6 Yard Illumination & Walkway

In the yard area where shunting activities/rake inspection are carried out during night hours needs to be illuminated properly. Particularly in Wagon Tippler area where coupling/decoupling of each wagon is done round the clock, there must be proper yard lighting to enable working persons to do the activity safely.

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Also, walkway along track is being made available where coupling/decoupling, shunting, TXR are carried out for movement of crew, to ease the movement.

5.7 Visit of MGR/FT System

1. Inspection of MGR/FT system by senior officials of station is to be done as per IOM ref- CC/FT/2020, Dtd 20.06.2020, issued by ED (FM & FT) (copy attached as **Annexure-IV**)
2. Record of visit along with observation, if any, is to be maintained at MGR/FT main C/R.
3. All staff in MGR Operation Shift duty to be assigned monthly once rota duty for Monitoring Unloading activity at WT / TH in general shift or MGR/FT Track circuit visit for better familiarization of ground activities involved in system operation and record of same with observations, if any, to be maintained.

5.8 Breath Analyzer

Breath analyzer - one fixed as per RDSO specification is provided at control room where SM/ASM/LD/ALD/Shunt-man/Points man report for the duty.

Breath analysis test are taken twice, one at the time of signing-in and second at the time of signing-out from the duty [Night Shift]. Attendance of the SM/ASM/LD/ALD/Shunt-man/Points man linked with the breath analyzer test.

Random checking by breath analyzer to be taken particularly during night hours at all control rooms /Track Hopper / Wagon Tippler etc. with the portable breath analyzer

5.9 Tools & Plants

Control Room - Red & Green hand flags, red banner flag, tri-colour (red/green/white) torch, torch (search/dragon light), first aid box, umbrella, C-clamp, skids, joggled fish plate. LD/ALD – Red & Green hand flags, tri-colour torch (red/green/white), duster cloth. Loco – skids, fire extinguisher, knuckle pin, hose pipe, hammer, C-clamp. LC Gate - Red & Green hand flags, red banner flag, tri-colour torch (red/green/white), pickaxe/crowbar.

5.10 Yard management/facilities

Identification board for all turnouts (point machines & Hand points) & line are provided so that LD/ALD can easily identify from the Loco. Board for turnout are provided close to the point machine/hand ever and for line close to the “FM” mark. Lettering of these board are done with fluorescent paint so that they are visible during night also

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5.11 Training & Competency Development

MGR/FT operation is specialized nature of job, which requires training for familiarization about the know-how of the system. Hence, time-to-time training to the concerned persons in form of safety induction training, job specific training, refresher training, fire-safety & training through IR/Railway- PSU Institutes/Personnel are imparted and record of same is to be maintained.

6.0 OPERATION OF TRAIN MOVEMENTS:

6.1. General

- 6.1.1. Control room/ stations mentioned in this document refers to S&T control building with ASM panel where SM/ASM sits and controls the movement of rail traffic.
- 6.1.2. SM/ASM at each control station will work under the guidance/instruction of main/centralized traffic controller – TH Control room.
- 6.1.3. All movements of trains are controlled by signals and point operation from panel. However, for operation of hand points / manual operation at locations / yard shall be controlled by clear working instructions as approved by MGR head.
- 6.1.4. In case of point / signal failure, the working procedures are framed and implemented by station MGR. All these working instructions / procedures shall be subjected to technical audits/ safety audits of station.
- 6.1.5. MGR operation at locations / yard will be done by SM/ASM with the help of points man / shunt- man etc.

6.2. System of Train Working

Over a period of time and with the evolvement of technology, different types of systems have evolved for the movement of Train between stations safely and efficiently. One train at a time in one block section working is followed at the time known as Absolute Block Working system for train operation. Unlike a road vehicle, train has huge inertia hence requires a much longer distance to break so, it cannot be expected from Loco Driver to stop if there is obstacle.

Details are given at **Annexure – III**

6.3 Caution order and permit

Caution order / look out notice for the activities planned by maintenance is to be submitted to main control room one day prior to the scheduled day of work. Further, Shift in-charge/Traffic-controller of main control room will issue caution order / look out notice to LD/ALP and same is to be acknowledged by LD/ALP for the compliance. One copy to be issued to LD/ALD and the other copy to be kept by control room for record. Also Caution / look out notice is to be

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shared by NTPC Shift I/c with IR Station Master in Rail fed stations for safety of the Gang man working at site and as information to Railway Drivers. In case of work required to be executed in Block permit, the request is to be submitted in main control room for written permission (hard copy/SAP) from shift In-charge, after getting permission of shift In charge, work at site is to be started only after exchange of Private no. between Shift I/c and Maintenance representative at site.

6.4 Train movement between Stations

Loaded rake movement

Loaded rakes Receiving to Crossing Station from Duvvada.

Load rakes on arrival at Duvvada, is received to NTPC siding, after exchanging of Private Numbers between Crossing Station master & Duvvada station master. Line clear will be granted to Duvvada with Token Less Cooperative Block Instrument. The line diagram & Points position for admitting load shown in the fig (1) & table (1).

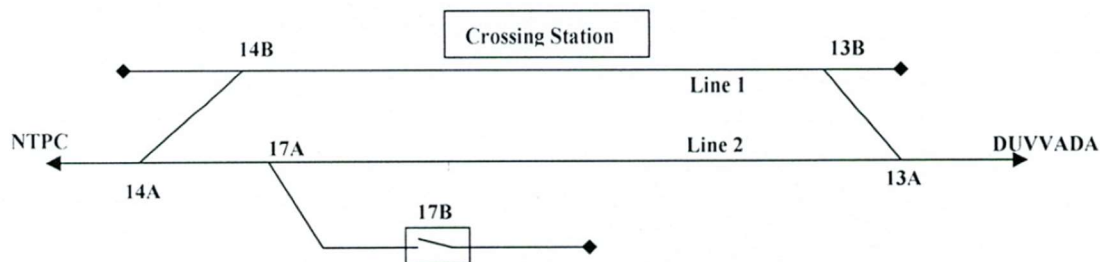


Figure (1)

Table (1)

Crossing Station Load Rake admitting/stabling from Duvvada			
Line	Condition	Point No	Position
Line No: 1	Line No: 1 clear. Line Clear to Duvvada.	13A	Reverse
		13B	Reverse
Line No:2	Line No: 2 clear. Point no: 17A should be normal. Line Clear to Duvvada.	13A	Normal
		14A	Normal

Upon load rake arrival from Duvvada, Line clear will be closed to Duvvada with Token Less Co-operative Block Instrument.

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Load rakes Dispatching from Crossing Station to R&D yard:

Load rakes on arrival at Crossing Station is dispatched to NTPC R&D yard, after exchanging of Private Numbers between LC gates 2, 3 & with R&D cabin master. Line clear will be obtained from R&D cabin with Token Less Cooperative Block Instrument. The line diagram & points operation for dispatching load rake are shown in the fig (1) & table (2). All load rakes, after dispatch from crossing station, on approach of Level Crossings (before 600 metres) will blow horn by seeing the W/L Sign Boards subsequent to which the LC gate will be closed. (One person per shift is manned at LC gates 2 and 3 each). All enroute speed restrictions on track will be followed by Railway Loco Pilots. Details of cautions applicable in NTPC Railway siding are provided to Duvvada Station Manager by MGR group.

Table (2):

Crossing Station Load Rake Dispatching to R&D yard			
Line	Condition	Point No	Position
Line No: 1	Line Clear from R&D All signal Clear.	14B 14A	Reverse Reverse
Line No:2	Block Line Clear from R&D All signal Clear.	14A 14B	Normal Normal

Upon load rake arrival at R&D, Line clear will be closed by R&D Cabin with Token Less Co-operative Block Instrument.

Loaded rakes Receiving to R&D Yard from Crossing Station.

Load rake on arrival at starter signal of R&D yard will check for signal clear & Home Signal cum

Route Indicator and proceed to the yard. The line diagram & points operation for dispatching load are shown in the fig (2) & table (3)

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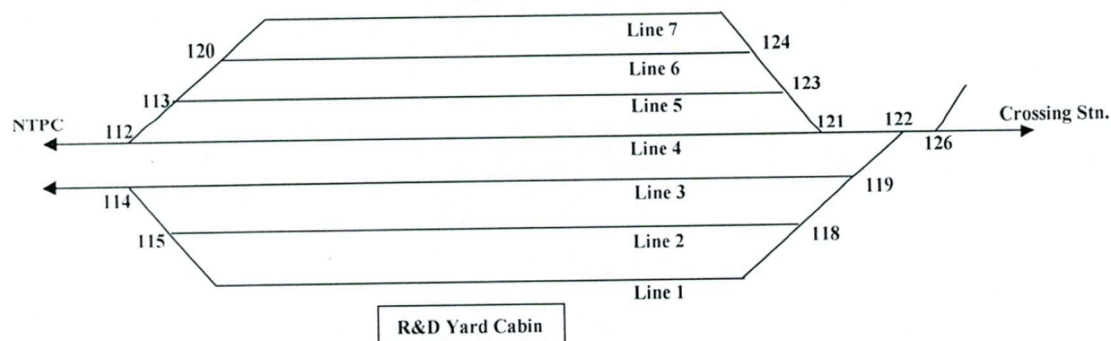


Figure (2)

Table (3)

R&D Yard Load Admitting/Stabling			
Line	Condition	Point No	Position
Line No: 1	Line no: 1 clear. Point no: 114, 115 should be reverse	126 122, 119, 118	Normal Reverse
Line No:2	Line no: 2 clear. Point no: 114, should be reverse 115 Normal	126, 118 122, 119	Normal Reverse
Line No: 3	Line no: 3 clear. Point no: 114, 115 should be Normal	126, 119 122	Normal Reverse
Line No: 4	Point No: 112 Normal	126, 122, 121	Normal
Line No: 5	Point No: 113 Normal 112 Reverse	126, 122, 123 121	Normal Reverse
Line No: 6	Point No: 120 Normal 113 Reverse 112 Reverse	126, 122, 124 121, 123	Normal Reverse
Line No: 7	Point No: 120 Reverse 112, 113 Reverse	126, 122 121, 123, 124	Normal Reverse

Upon load rake arrival at R&D yard line clear will be closed by R&D Cabin with Token Less Cooperative Block Instrument.

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Loaded rakes Dispatching to TH/WT from R&D Yard:

Load rakes on arrival at R&D Yard, is dispatched to TH/WT, after exchanging of Private Numbers between R&D Cabin & TH and with intimation to IMBW & CISF R&D Gate Control Room.

Slot will be granted by TH/WT Control Room to R&D with RRI system. Based on the Slot allotted, R&D will clear the signals and send the rake to TH/WT. The line diagram & Points operation for Dispatching load are shown in the fig (3) & fig (4). Normally, Load rakes are entered to the plant through Gate no-A.

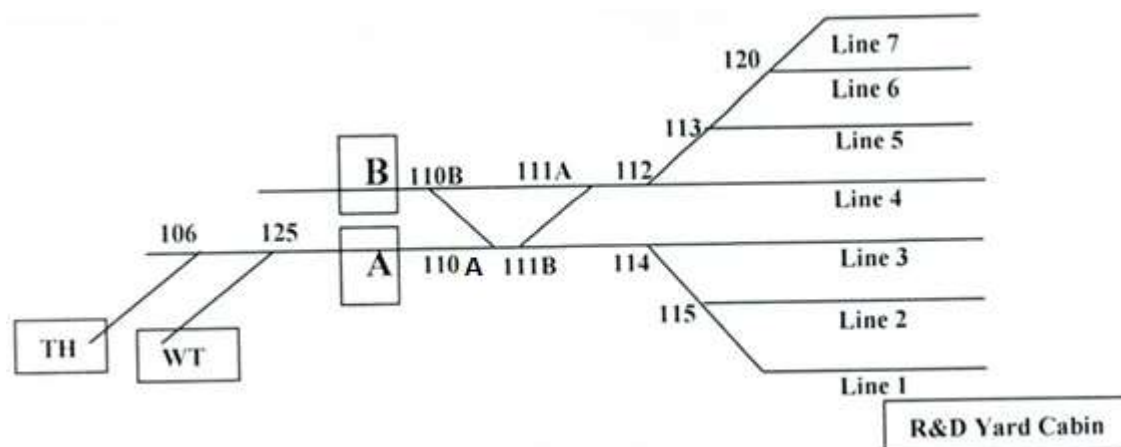


Figure (3)

Table (4)

R&D Yard Load Dispatching to TH/WT			
Line	Condition	Point No	Position
Line No: 1	Slot Granted from TH/WT	114, 115 111, 110	Reverse Normal
Line No:2	Slot Granted from TH/WT	114 115, 111, 110	Reverse Normal
Line No: 3	Slot Granted from TH/WT	114, 111, 110	Normal
Line No: 4	Slot Granted from TH/WT	112, 110 111	Normal Reverse
Line No: 5	Slot Granted from TH/WT	113, 110 112, 111	Normal Reverse
Line No: 6	Slot Granted from TH/WT	112, 113, 111 120, 110	Reverse Normal

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Line No: 7	Slot Granted from TH/WT	112, 113, 120, 111 110	Reverse Normal
Dispatching to TH		125 106	Normal Reverse
Dispatching to WT		125	Reverse

Loaded rakes receiving to TH/WT from R&D Yard:

Load rake after dispatched from R&D passing on through IMWB and Gate-A can be taken to TH or WT. The Line diagram & Points operation for Dispatching load are shown in the fig (4) & table (5)

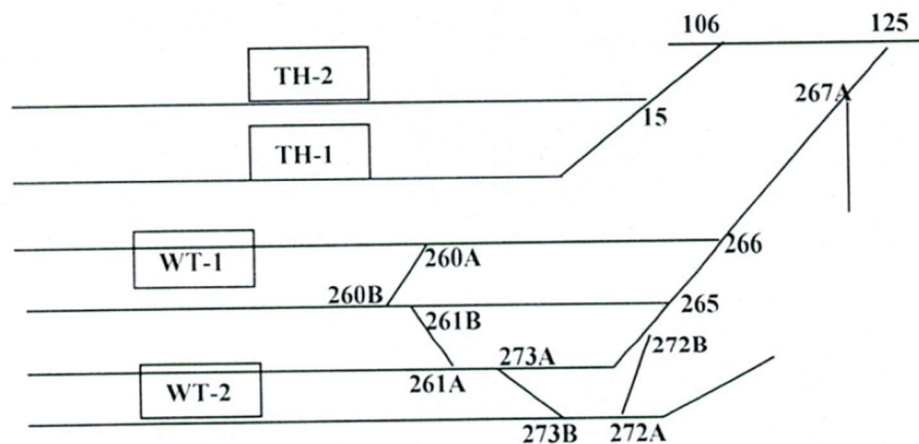


Figure (4)

Table (5)

TH/WT Load Admitting			
Line	Condition	Point No	Position
TH:1	TH:1 Clear	15	Normal
TH:2	TH:1 Clear	15	Reverse
WT:1	WT:1 Clear	267 266 260*	Normal Reverse

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WT:2	WT:2 Clear	267, 266, 265, 272B, 273A 261*	Normal Normal
Ash Silo Lines	Slot Granted form TH/WT	267, 266, 265 272B, 272A 273B, 273A	Normal Reverse Normal

The BOBRN load rake on arrival at Track Hopper is admitted & placed for unloading of the first 20 Wagons with the help of compressed air to the cylinders fitted on the Railway wagons.

The BOXN load rake on arrival at Wagon Tippler is admitted & stabled in reach of Side Arm Charger (SAC) for detaching of LOCO, uncoupling of wagon brake pipes, and release of brakes for pulling of rake/wagons by Side Arm Charger(SAC), for placement on Wagon Tippler table, for unloading/tipling.

(*) – Point Number 260A & 260B will be kept Reverse for engine escape and will be kept Normal for back push of rake for WT-1.

Point Number 261A & 261B will be kept Reverse for engine escape and will be kept Normal for back push of rake for WT-2.

Engine movement:

Light Engine escaping to R&D after placing load at WT:

Upon placing of the Load rakes at WT the Light engines are sent to R&D after exchanging of Private Numbers between R&D cabin & TH Control Room and with an intimation to CISF R&D Gate Control Room.

First Slot will be granted by WT Control Room for Light engines up to signals S200/S201. Parallely Second Slot is granted by R&D for empty rakes to move beyond signals S200/S201. The line diagram & Points operation for Dispatching Light engines are shown in the fig. (4), (5) & table (6). Light engines are sent to R&D through either Gate no-A or Gate no-B based on the requirement/convenience at R&D Yard.

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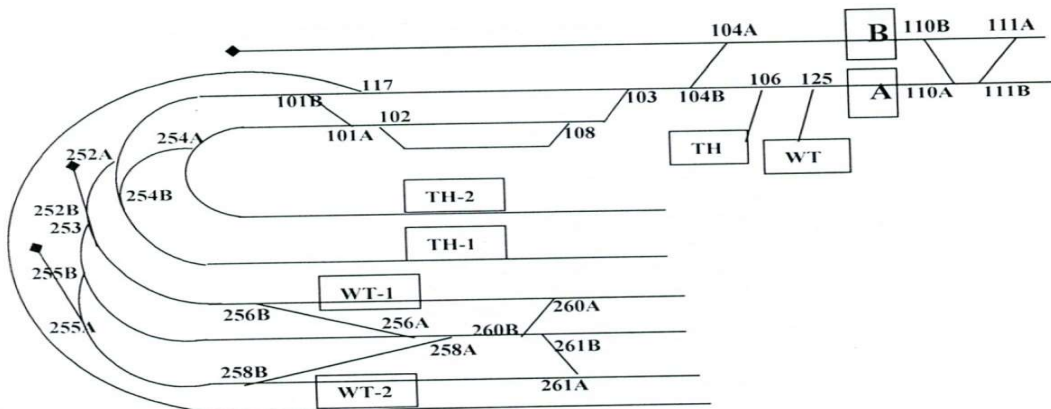


Figure (5)

Table (6)

Engine escaping to R&D after placing load at WT			
Line	Condition	Point No	Position
WT: 1		266	Reverse
		266	Normal
		265	Reverse
		260B, 261B	Normal
		258A, 256A	Normal
		255B	Normal
		253	Reverse
		252B, 252A	Reverse
		101B, 103	Normal
		104B, 104A	Reverse
WT: 2		272B	Normal
		265	Normal
		266	Normal
		266	Normal
		265	Reverse
		260B, 261B	Normal
		258A, 256A	Normal
		255B	Normal
		253	Reverse
		252B, 252A	Reverse
		101B, 103	Normal

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		104B, 104A	Reverse
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Light Engines to WT for Empty Drawl/ASH Line Load Drawl:

Upon unloading completion of the Load rake at WT the Light engines are sent to WT from R&D after exchanging of Private Numbers between R&D cabin & WT and with an intimation to CISF R&D Gate Control Room.

The Light engines are sent to WT/Ash Line from R&D for empty/Ash Load drawl. The Line diagram & Points operation for dispatching Light engines are shown in the fig (4), (5) & table (7). Light engines are sent to R&D through either Gate no- A or Gate no- B based on the convenience and engine on the line no at R&D Yard.

Table (7)

Light Engines to WT Empty Drawl/ASH Drawl			
Line	Condition	Point No	Position
WT:1	Through Gate-A	125, 106, 104B 103, 117, 101B 252B, 252A 253, 256B	Normal Normal Reverse Normal
WT:2	Through Gate-A	125, 106, 104B 103, 117, 101B 252B, 252A, 253 255B, 255A 258B	Normal Normal Reverse Reverse Normal
Ash Silo Lines	Through Gate-A	125, 106 104B, 103 117	Normal Normal Reverse
WT:1	Through Gate-B	104A, 104B 103, 117, 101 252B, 252A 253, 256B	Reverse Normal Reverse Normal
WT:2	Through Gate-B	104A, 104B 103, 117, 101 252B, 252A 253, 255B, 255A 258B	Reverse Normal Reverse Reverse Normal

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Ash Silo Lines	Through Gate-B	104A, 104B 103 117	Reverse Normal Reverse
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EMPTY RAKE MOVEMENT

Empty Rake dispatching to R&D from TH/WT:

Upon unloading completion, Empty rake and Light engines are attached coupled & after charging of brake pipe pressure, the Empty rake will proceed from TH/WT to R&D.

Empty rake is dispatched to R&D after exchanging of Private Numbers between R&D cabin & TH Control Room with intimation to IMWB & CISF R&D Gate Control Room.

Empty rake from TH – To dispatch empty rake from TH, Main Signal S1 for TH1 or Main Signal S2 for TH2 is taken off from TH Control Room, for which permissive Slot to be granted by WT Control Room, so that empty rake will move upto main signal S200/S201. Second Slot is granted by R&D Control Room for both the empty rakes to move beyond signals S200/S201.

Empty from WT – To dispatch empty rake from WT, Shunt Signal SH210 for WT1 or Shunt Signal SH208 for WT2 is taken off from WT Control Room, so that empty rake will move up to main signal S200/S201. Second Slot is granted by R&D Control Room for both the empty rakes to move beyond signals S200/S201.

The line diagram & Points operation for dispatching empty rakes are shown in fig. (4), (5) & table (8). Normally, empty rakes are sent to R&D through Gate No-B.

Table (8)

TH/WT Empty Dispatching			
Line	Condition	Point No	Position
TH:1	Through Gate-B, Slot Granted from TH & WT	254B, 252A 101B, 103, 117 104B, 104A	Normal Normal Reverse
TH:2	Through Gate-B, Slot Granted from TH & WT	254A, 101A, 102 108, 103 104B, 104A	Normal Reverse Reverse

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WT:1	Through Gate-B, Slot Granted from WT	256B, 253 252B, 252A 101B, 103, 117 104B, 104A	Normal Reverse Normal Reverse
WT:2	Through Gate-B, Slot Granted from WT	258B 255A, 255B 253 252A 101B, 103, 117 104B, 104A	Normal Reverse Reverse Reverse Normal Reverse
Ash Silo Lines	Through Gate-B	117 103 104B, 104A	Reverse Normal Reverse

Empty rake receiving at R&D Yard

Empty rake is dispatched to R&D after exchanging of Private Numbers between R&D cabin & TH Control Room with intimation to IMWB & CISF R&D Gate Control Room.

Empty from TH – To dispatch empty rake from TH, Main Signal S1 for TH1 or Main Signal S2 for TH2 is taken off from TH Control Room, for which permissive Slot to be granted by WT Control Room, so that empty rake will move upto main signal S200/S201. Second Slot is granted by R&D Control Room for both the empty rakes to move beyond signals S200/S201.

Empty from WT – To dispatch empty rake from WT, Shunt Signal SH210 for WT1 or Shunt Signal SH208 for WT2 is taken off from WT Control Room, so that empty rake will move upto main signal S200/S201. Second Slot is granted by R&D Control Room for both the empty rakes to move beyond signals S200/S201.

The line diagram & Points operation for dispatching empty rakes are shown in fig. (4), (5) & table (9). Normally Coal Empty rakes are sent to R&D through Gate No-B.

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Table (9):

Empty rake receiving at R&D Yard			
Line	Condition	Point No	Position
Line No: 1	Through Gate-B, Slot Granted from TH/WT Point Number: 118, 119, 122 - Reverse	110B, 110A 111B 114 115	Reverse Normal Reverse Reverse
Line No: 2	Through Gate-B, Slot Granted from TH/WT Point Number: 118, 119, 122 - Reverse	110B, 110A 111B 114 115	Reverse Normal Reverse Normal
Line No: 3	Through Gate-B, Slot Granted from TH/WT Point Number: 122 - Reverse	110B, 110A 111B 114 119	Reverse Normal Normal Normal
Line No: 4	Through Gate-B, Slot Granted from TH/WT Point Number: 121, 122, 126 - Normal	110B 111A 112, 113	Normal Normal Normal
Line No: 5	Through Gate-B, Slot Granted from TH/WT Point Number: 121 - Reverse 120, 122, 123, 124 - Normal	110B 111A 112 113	Normal Normal Reverse Normal
Line No: 6	Through Gate-B, Slot Granted from TH/WT Point Number: 121, 123 – Reverse, 120, 122, 124 - Normal	110B 111A 112 113	Normal Normal Reverse Reverse
Line No: 7	Through Gate-B, Slot Granted from TH/WT Point Number: 120, 121, 123, 124 – Reverse, 122 - Normal	110B 111A 112 113	Normal Normal Reverse Reverse

R&D Yard Empty rakes dispatching to crossing station.

Empty rakes on arrival at R&D Yard, is dispatched to Crossing Station, after exchanging of Private Numbers between LC gates 02,03 & with Crossing Station Line clear will be obtained from Crossing Station with Token Less Cooperative Block Instrument.

The Line Diagram & Points operation for Dispatching Empty rakes to R&D are shown in the

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fig. (2) & table (10).

Table (10)

R&D Yard Empty Dispatching			
Line	Condition	Point No	Position
Line No: 1	Line no: 1 clear. Point no: 114, 115 Reverse	126 122, 119, 118	Normal Reverse
Line No:2	Line no: 2 clear. Point no: 114, reverse. 115 Normal	126 122, 119 118	Normal Reverse Normal
Line No: 3	Line no: 3 clear. Point no:114, should be Normal	126 122 119	Normal Reverse Normal
Line No: 4	Point No: 112 Normal	122 121 126	Normal Normal Normal
Line No: 5	Point No: 113 Normal 112 Reverse	122 121 123 126	Normal Reverse Normal Normal
Line No: 6	Point No: 120 Normal 113 Reverse 112 Reverse	122 121, 123 124 126	Normal Reverse Normal Normal
Line No: 7	Point No:120 Reverse 113 Reverse 112 Reverse	122 121, 123 124 126	Normal Reverse Reverse Normal

Crossing station empty admitting/stabling.

The Line Diagram & Points operation for admitting/stabling Empty Rake are shown in the fig. (1) & Table (11).

Table (11)

Crossing Station Empty admitting/stabling			
Line	Condition	Point No	Position
Line No: 1	Line No: 1 clear.	14A 14B	Reverse Reverse

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Line No:2	Line No: 2 clear Point no: 17A should be Normal	13A 14A	Normal Normal
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Upon Empty rake arrival from R&D line clear will be closed to R&D with Token Less Cooperative Block Instrument.

Empty rake Dispatch to Duvvada from Crossing Station.

Empty rakes on arrival at Crossing Station, is Dispatched to Duvvada, after exchanging of Private Numbers between Crossing Station Master & Duvvada Station Master. Line clear will be granted by Duvvada with Token Less Cooperative Block Instrument. The line Diagram & Points operation for Dispatching load are shown in the fig. (1) & table (12).

Table (12)

Crossing Station Empty Dispatching to Duvvada			
Line	Condition	Point No	Position
Line No: 1	Block Line Clear from Duvvada All signal Clear.	13A 13B	Reverse Reverse
Line No:2	Block Line Clear from Duvvada All signal Clear. Normal	13A	Normal

Upon Empty rake arrival from Crossing Station Line clear will be closed Crossing Station with Token Less Cooperative Block Instrument by Duvvada (Rly).

6.5 Shunting Movement

Not applicable at Simhadri as Simhadri is not having any Rolling Stock. All the Train Movements are done by Indian Railways Engines which are bringing the trains to Simhadri.

Shunting staff composition

Shunting operations are limited to inside STDV Yard. Loco Pilot and Points man will do all shunting movements with the information from Station Master for any Brake van and other rake/Wagon/Loco movements by observing signalling system (Shunt Signals).

During shunting of rake, one points man shall be on the Loco and one points man with walkie-talkie and flag/tri-colour torch signal should be maintained for safety at near point/last wagons.

6.6 Operations in case of enroute loco failure / rail stalling

Not Applicable at Simhadri as Simhadri is not having any Rolling Stock.

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6.7 Operations in case if signal failure:

In case of Signal failure, a Pilot in Memo (written authority) is issued by the concerned Station Master to the Loco Pilot and TPM will assist the rake movement by setting the Route (if Points are also not operable from panel) to the designated Unloading Point i.e., WT or TH.

Written pilot in memo:

It is issued for rake movement in case of emergency situations like signal failure communication failure.

6.8. Operation in case of Motorized point failure / hand point

In case of failure of motorized point and in hand point zone, point will be set properly by points man /shunt-man and locked by C-clamp before movement.

6.9. Trolley/RRV Movement

Movement of trolley/dip lory/ rail dolly/RRV in the block section will be treated as train and carried out with absolute block working. However, trolley/dip lory/ rail dolly/RRV is supposed to cover partial length of block in that case trolley/dip lory/ rail dolly in-charge will give clearance to the respective control room by exchanging private number after clearing the track and then only train movement is to be carried out.

Alternatively, if the movement in tracks are unidirectional (two separate tracks for empty & load respectively) then movement of trolley/dip lory/ rail dolly/RRV done after departure of train by maintaining time gap of at least 15 minutes and physical separation of 2 KM with caution to trolley in-charge. However, for such movement private no. exchanged with concerned control station and control station shall ensure that other movement on the track is not permitted till the line clear has been given by the in-charge of trolley/dip lory/ rail dolly, further control station will inform such movement to the control station in advance and Loco Driver of train ahead. IOM Dtd 22.06.2023 "Guidelines for trolley operation & track maintenance activities" referred for details.

6.10. Single Line Operation

Single line movement is to be done if it is implemented in S&T system & system is functioning properly or only in the case of exigencies.

For doing any single line movement which is not designed in the system, MGR Operation In-charge, traffic supervisor & SM/ASM of stations at both ends will strategize the detailed movement plan in writing and get it approved from O&M Head.

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While performing single line movement in-case of outage of one-track/failure of Loco etc. & S&T system is not designed for the same, then following precautions needs to be taken care in addition to rake movement mentioned above –

If the S&T system is not designed for bi-directional movement then movement in one of the directions cannot be done using the block instruments & signals. Movement on the available single track to be done following the absolute block working only following the private no. exchange and authority to proceed. A separate private no. book with private no. on both end of rows to be prepared and provided at both stations. Private no at beginning of the row to be used for sending rail vehicle into block and private no. at end of the same row to be used for completing the movement i.e., after complete arrival of rail vehicle, SM/ASM of receiving station will exchange with SM/ASM of sending station to complete the movement. In this way it can be ensured that when a row of private no book has entries against both private nos. of a row then track is not occupied.

Before granting line clear to SM/ASM of sending station, SM/ASM of receiving station will ensure complete arrival of previous Rake/Loco. For each movement, after ensuring complete arrival of a train / rail vehicle, SM/ASM of receiving station will inform the arrival to the SM/ASM of sending station by exchanging the private no. (at end of row), LV no, time of arrival etc. and the track will be close the line for traffic.

Further two separate colour pens may be used to enter the private no book for each direction of movement, to make it more clear.

Movement of trolley/dip lorry/ rail dolly/RRV to be avoided during the period of single line movement in the healthy track. If necessary, this movement to be done by following the absolute block working only.

After the movement of rail vehicle in healthy track, setting of points at both stations end to be such that in case of overrun by another vehicle it should be not enter the healthy track. For this purpose, portions of unhealthy track after the points at both stations to be kept as overrun track and after this portion track may be obstructed/opened to avoid further entry (act as derailment switch).

6.11. Sampling

1. Regarding sampling of coal from wagon OGN “COS-ISO-00-OGN/OPS/SYST/058, SAFE WORK PRACTICES IN ELECTRIFIED TRACK SECTION” and “COS-ISO-00-OGN/OPS/CHEM/006, GUIDELINES ON SAMPLING AND ANALYSIS OF COAL” are referred.
2. Sampling from wagon is to be done by providing sampling platform and jumping from one wagon to another is avoided.

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6.12. Rake Unloading

Rake unloading is very sensitive from the safety perspective hence, clear written instructions are prepared by the station for unloading of rakes & displayed at suitable locations. Some basic requirements are given here.

1. Hooter/Siren to warn workers of rake movement. This hooter/siren is blown before rake placement and before each rake movement to warn worker.
2. Unloading signals – Unloading signals are installed inside TH and at suitable locations for LD to given visual instruction/status of the unloading activity. Before any rake movement, unloading supervisor & sampling supervisor (i.e. parties involved in unloading) will give their clearance (interlocked) for the movement and only after that these signals can be operated.
3. Unloading supervisor – Unloading supervisor is available for unloading activity separately for each place. He will continuously monitor the unloading activity and coordinate with the control rooms. Only after ensuring that all engaged workers are withdrawn from the unloading activity & track is clear for movement, he will give clearance to the control room for the movement of rake.
4. There are a dedicated communication means for supervisors to be able to clearly communicate with LD/ Control Room/ WT operator through Walkie-talkies.
5. Unloading at WT is done by following the guidelines defined in OGN “Common Standard Operating Procedures for Wagon Tippler Operations, COS-ISO-00-OGN/OPS/SYST/037.”

6.13. Parking/Stabilizing

Whenever parking/Stabilizing Loco/Wagon at least two nos of skids are applied in the opposite direction of wheel. If it is required to park these vehicles for a long time, then hand brake are applied and wagon is chained to rail in addition to application of skids.

If it is required to park a rake, then hand brake will be applied on first five & last five wagons and skids are applied.

In no case LD/ALD will leave the running (start condition) loco parked.

6.14. Standard Operating Procedures (SOP)

SOPs for all activities like shunting, sampling, single line operation, unloading of rakes, OHE isolation, rake parting & placement etc. are prepared and approved by MGR I/c. SOPs are prepared in bilingual and displayed at relevant locations. SOPs are acknowledged by all concerned involved in the activity and is discussed/explained during pep-talk.

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7.0 Safety Procedures / Emergency Response Plan

7.1. Safety Procedures

Standard safety working procedure / instructions are developed in case fire, thunderstorm, power cut (blackout) and water logging. These are displayed in each MGR / crossing station control rooms and at all important visible locations. These instructions are made in local language also apart from English.

Safety precautions are defined by station for movement of trains, shunting of trains, movement of light loco etc. These are approved by station MGR and made available to all the concerned. This is also part of safety training / safety meetings at the site.

Safety precautions for trains, loco and detached wagons are also framed, and these are conveyed to all concerns clearly.

- These contain, application of hand brakes, use of skids, use of reverser handle, loco parking, detached wagon parking etc.

Emergency response plan for MGR/FT system of Simhadri station covering all the possible incidents/accidents and responsibilities of each department for earliest normalization of system, is prepared, routed through FES/P&S, HR, Medical, Safety and approved by HoP.

O&M head / station head reviews the exceptions / near miss- incidents on monthly basis.

Disaster Management Plan is attached as Annexure-X

7.2. In case of Accident / Derailments.

Standing preliminary investigation team needs to be constituted at all stations by HoP for enquiry of accidents / derailment comprising of member from Safety, Rolling Stock, Track & MGR/FT Operation.

The responsibility of the preliminary investigation team is to take all the readings of track, involved wagons parameters and collection of substantial evidence which shall help the accident enquiry team to reach conclusion of reasons of derailment. Further, detailed enquiry report including cause and remedial action to avoid repetition of the same incidents is to be shared by MGR/FT In-charge with all stations.

FT system circular - 01/2012 dtd 31.07.2012 on MGR/FT accidents – derailment reporting, restoration and enquiry system issued by ED (FT) & further re-issued by ED (OS) dtd. 14.12.22, is to be followed (**Annexure-V**). Responsibility of sharing the Flash Report of

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derailment / accident/ incident in the standard format with HoP, RED, ED (OS), ED (FM/FT) and Dir (O) will be of MGR/FT-O In-charge within 1 Hr of the incident.

While restoration work is being carried out, all standard IR/NTPC safety norms need to be followed. Safe operation of all Re-Railing equipment, cranes etc. are to be ensured by station MGR head.

After completion of restoration work, proper track fitness is to be issued by the Track Maintenance section before Loco rolling.

A. Reporting of Incident:

- a. Flash report of incident (through fax/email/SMS/WhatsApp) should consist of:
 - Location & sequence of event
 - Rolling stock involved
 - Injury if any
 - Likely reason
 - Approx. Loss
 - Anticipated time for restoration-if applicable.
- b. Report to be flashed to RED, ED(OS), ED(FT), CGM CC FM, D(Fuel) & D(O) within 01 hour of incident.
- c. Type of Incident:
 - i. Derailment with obvious reason such as:
 - Riding over the coal at Track Hopper/WT or at Track.
 - Wagon door opening
 - Derailment which did not involve loss of rolling stock.
 - ii. Derailment/accident involving loss of rolling stock/ MGR assets / Human life / run over.
- d. Detailed records of all derailments should be maintained by Site MGR in a Derailment Register, which will be subjected to inspection during Technical Audits.

B. Start of restoration work

- a. Head of station should form a standing Preliminary Investigation Team comprising of members from Rolling Stock, Track, MGR-Operation and Safety. This team shall be in place and should reach to accident site as soon as possible and shall take up the activities as under:
 - For incident mentioned in A-c(i), the restoration work shall start immediately after investigation by any two members of Preliminary Investigation Team.
 - For incidents mentioned in A-c(ii), the restoration work should be started only after the clearance subsequent to critical inspection and investigation by Preliminary Investigation Team.
- b. Preparation of site sketch depicting the direction of movement, position of

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derailed/capsized rolling stock, identifying & displaying the point of drop/mount, displaying the markings on sleepers track etc. & assessment of the probable primary reasons of incident should be done by Preliminary Investigation Team.

- Recording of track parameters of 30 meter track on both side of the accident site, Measurement of Wagon & Loco wheel profile of the involved rolling stock wherever possible, their coupler conditions, S&T system conditions near site and recording of brief statement of the driver/guard/station staff/points-man etc.
- c. After assessment, the Investigating Team shall allow the site staff/station FM to start restoration work and shall prepare a report which shall be submitted to the Enquiry Committee for subsequent investigations.
- d. Site FM should plan & mobilize for restoring of coal movement in such an event which may involve single line operation / movement in reverse direction / slow speed cautioned movement etc. keeping safety practices in mind.
- e. Site FM should restore the system to normalized state & traffic operations to be restored to normal operations.
- f. Site should inform the system normalization to CC OS, CC FT & Regional Office.

C. Constitution of Enquiry Committee & Reports on Enquiry:

- a. In line with Corp. Operation directive (COS-ISO-00-OD/GEN/004 Rev. No.:2, October,2005); the Enquiry Committee should be constituted by Competent Authority (C.A.) within 24 hrs of reporting of incident. The responsibility of formation of enquiry committee is as under:
 - i. Corporate FT Dept.: where C.A. is D(O)
 - ii. Regional OS/FM: where C.A. is RED
 - iii. Station MGR/FM: where C.A. is HOS
- b. Enquiry Committee should consist at least two members (from MGR stations/RHQ/CC) having sufficient working experience and knowledge of MGR & Railway systems in order to have qualitative investigation of the incidents. Enquiry Committee should inspect the site & start investigation as early as possible. The collected data and report of preliminary Investigation team should be given to Enquiry committee.
- c. Enquiry Committee should investigate / enquire on various technical, operation & safety aspects of incident and thereafter should submit the report to C.A. in the following format:
 - i. Preamble
 - ii. Committee constitution detail.
 - iii. Time schedule
 - iv. Enquiry Proceedings
 - Description of accident site (with photo, sketch etc.)
 - Detail of Track, S&T etc.
 - Details of documents & record checked.
 - Investigation & Committee observations.

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- Establishment of reason of accident
- Cost of damages.
- v. Recommendations / Remedial measures suggested.
- d. In case, the accidental incidents are repeated, at any station, an external independent agency should also be engaging in the investigation.

D. Submission of Report & follow-up

- a. Report to be submitted to the competent authority by the leader / Head of Enquiry committee.
- b. Regional OS/FM/ORT coordinator should monitor the implementation of enquiry recommendation after approval through various meetings such as monthly ORT.
- c. Regional ORT Coordinator shall forward the Enquiry Report to CC-FT for circulation of Recommendations of the Enquiry Commit other stations / uploading the same on Web page of Corporate-FM/MGR for avoidance of such incident by modifying the practices by other stations as per enquiry findings.
The Enquiry Closure Report to be submitted by Regional OS/FM, to CC-FT stating the reasons & responsibilities as per the findings of the Enquiry report.
- d. Station FM shall maintain and keep the records of all enquiries and the enquiry reports after implementation of recommendations and shall be the custodian of the same.

Implementation: Head of Station (HOS) shall be responsible for implementation of Enquiry Committee report recommendations

8.0 LMI preparation

Based on the above guidelines as issued by Corporate OS, Simhadri station has prepared LMI (Local Management Instructions) with all systems, procedures and responsibilities. LMI is approved by HOP & is due for review every two years or as per site requirement to incorporate any issued guidelines. The copy of approved LMI is to be acknowledged by all concerned employee of MGR/FT for compliance.

These LMIs shall be subjected to audit during technical audit and LMI audit of the station vis-a-vis actual followed practices and the exceptions thus noticed shall need to be corrected and accordingly LMI to be revised.

9.0 Abbreviations

IR – Indian Railway

SM – Station Master

ASM – Assistant Station Master

LD/LP – Loco Driver / Loco Pilot

ALD/ALP – Assistant Loco Driver / Assistant Loco Pilot

RRV - Rail cum Road Vehicle

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10.0 Reference:

1. Indian Railway General & Subsidiary Rules
2. Spec RDSO/2015/EL/SPEC/0119 (Rev 1) "Fuel Cell Sensor Based Breath Alcohol Analyser for General Purpose Use on Indian Railways"
3. COS-ISO-00-OD/GEN/004 "System for Reporting arrangements and Investigation of Accidents, Dangerous Occurrences and Major Incidents in a Power Station"
4. OGN - COS-ISO-00-OGN/OPS/SYST/058, SAFE WORK PRACTICES IN ELECTRIFIED TRACK SECTION
5. OGN - COS-ISO-00-OGN/OPS/CHEM/006, GUIDELINES ON SAMPLING AND ANALYSIS OF COAL

11.0 Associated documents

- Annexure - I: List of Records to be maintained at MGR Control Room
- Annexure - II: Operation System Check List
- Annexure - III: System of Train Working
- Annexure - IV: IOM "MGR/FT system visit by higher officials"
- Annexure - V: IOM "FT circular for accident/derailment reporting"
- Annexure - VI: IOM "Compliance of recommendations for safety and reliability in FT/MGR area"
- Annexure - VII: IOM "Supervision of Rail Operation by Heads of MGR/FT"
- Annexure - VIII: Do's & Don'ts for VHF equipment
- Annexure - IX: Monsoon & Summer Preparation
- Annexure - X: Disaster Management

Note – Circulars/IOMs related to MGR/FT can be accessed from Corporate OS (COS) intranet site "Corporate OS > OS Dashboard > Department > BoP > FT Group"

12.0 REVIEW

Review Responsibility	HOD-FC/FT
Review Frequency	On 02 yearly basis or as necessary
Review Approval	HOP

13.0 CHECKLIST AND ANNEXURES

The following is the check list and Annexures to be followed

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13.1. Annexure-I:

List of Records & Formats to be maintained at MGR Control Room

SI	Description	Action (Yes/No)	Remarks
1	Rake Movement Record	Yes	
2	Shift In-charge Logbook	Yes	
3	Authority to Proceed (Paper line Clear format)	Yes	
4	Private no. book Issue Record	Yes	
5	Private no. book	Yes	
6	Caution Order Format	Yes	
7	Loco log sheet	Not Applicable	
8	Route Observation Log sheet	Not Applicable	
9	Rake Unloading Record / Unloading Timing Sheet	Yes	
10	Rake Loading Record	Yes	
11	Shunting Memo	Yes	
12	Breath Analyzer Record	Yes	
13	Rake Brake Power Test Record	Not Applicable	
14	TXR examination of rake before dispatch (including brake continuity test)	Not Applicable	GDR test being done by Railway
15	Communication/Defect Register	Yes	
16	Enroute failure of Loco, wagons	Not Applicable	
17	Signal Failure Register	Yes	
18	Pep talk /Incident recall record	Yes	

Responsibility: SCE (TH) & DGM (FT-O)

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13.2. Annexure-II:

Operation System Check List

S.N o.	Check points	Status	Remark
1.	Whether prescribed medical examination of the loco Drivers and other operating staff is regularly carried out in time.	YES/NO	Loco driver – NA Operating staff - Yes
2.	Whether Long duty hours are defined / fixed at 10 Hrs. in line with the Railway working.	YES/NO	Not applicable
3.	Whether Railway drivers appointed have experience of operating WDS6/WDM2/ locomotives during their service in Railways.	YES/NO	Not applicable
4.	Whether Placement of (LV) board with buzzer during day hrs. and red lamp (tail lamp) with buzzer during night / thick and foggy weather is compulsorily followed.	YES/NO	Not applicable, However being followed as per Railway practices
5.	Whether all emergency contact no's are displayed in all important locations	YES/NO	YES
6.	Working instructions are prepared and are acknowledged in writing by all the staff associated with MGR train operation.	YES/NO	YES
7.	Whether station has prepared LMI, JSA, HIRA for rake operations and procedure of all type of rake movements, station working & shunting operations are specified and copy of the same is acknowledged and kept at all stations.	YES/NO	YES
8.	Whether station is imparting regular training of LMI, job specific trainings, sharing other NTPC plants/IR related accidents and safety aspects in operation to all operating staff	YES/NO	YES

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9.	Whether development/refresher training being provided to the operating staffs which include loco-pilots, ALPs, SM / ASM, points man/shunt man, technician etc, so that they should remain aware of safety requirements and safety consciousness. (As per CC IOM ref. no. CCFT/MGR/Safety/01 dtd. 2nd July'20)	YES/NO	Development/ refresher training being provided to the operating staffs which include SM / ASM, points man/shunt man, technician etc,
10.	Whether station working rules and station layout diagrams are available and displayed at MGR control room and at all Stations.	YES/NO	YES
11.	Whether the Train operational requirement of not permitting any rake to enter in the block section until and unless previous rake has cleared the block section and cleared the safe distance beyond the 1st stop signal of the next station is being followed.	YES/NO	YES
12.	Whether complete arrival / departure of the train is being ensured by the person granting permission through physical verification or other means of the last vehicle.	YES/NO	YES
13.	Whether private no. exchange system is in practice in addition to signals and verbal communication to ensure safety in rake & trolley operation.	YES/NO	YES
14.	Whether electronic breath analyzer to ascertain that the loco crew members are not under the influence of alcohol at the time of reporting for duty and on leaving after duty (signing 'ON' and signing 'OFF'), is installed in the central control room or the place where they report for duty. (As per one of the accident enquiry committee recommendations, the breath analyzer results are to be recorded in a register and counter-signed by the officer in-charge).	YES/NO	YES – Done for Retired Railway and Coal unloading staff

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15.	Whether disaster management plan is prepared and approved by CA for the MGR/FT system. (To be acknowledged by O&M staff)	YES/NO	YES
16.	Whether record of scheduled visit of MGR system (including night inspection) by senior officials / MGR executives as per CC IOM Dated 20/06/20 is being maintained.	YES/NO	YES
17.	Whether record of accidents/ incidents / derailments/ near miss / run-over in MGR/FT system is being maintained	YES/NO	No. of Near Miss : Yes No of Run-over: Yes No. of accident / derailments : Yes
18.	Whether System of providing Caution / Speed restrictions details to Loco drivers is being followed.	YES/NO	YES
19.	Whether any operating staff is working beyond the age of 65 Yrs.	YES/NO	NO
20.	Whether red/ green flags (day) & tricolor torches (night) are being used by pointsman/shunt-man during shunting.	YES/NO	YES
21.	Whether there is proper system of reporting of various defects to concerned maint. sections which are timely being acknowledged and rectified by those maintaining the sections.	YES/NO	YES
22.	Whether timings of all movements, which are components of rake cycle time, are being recorded and monitored.	YES/NO	YES
23.	Whether station is following system of preliminary investigation by committee comprising members from Operation, Track, Rolling Stock & Safety for all derailments/ accidents as per FT system circular no. 01/2012; dtd 31.07.2012	YES/NO	YES

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24.	Whether station is carrying out pep talks, awareness safety talks, incident recall, recorded audio message and safety pledges regularly with employees and workers	YES/NO	YES
25.	Whether station has prepared checklist of safety/ protection and standard operating Instructions including Do's/Don't in Hindi/English and most importantly in local language and displayed in all important location of MGR and Track Hopper area. The same should be reviewed time to time. The Clear visibility of these display at all times must be ensured.	YES/NO	YES
26.	Whether proper permit/ permissive system is in place for coal unloading staff to avoid rake movement during coal unloading at track hopper and provision of hooter warning exist for any rake movement/placement before giving clearance to the loco driver.	YES/NO	YES
27.	Whether station has Installed / Displayed caution signs/ warnings/ important instructions against electric hazard near OHE lines to improve awareness among the MGR/FT, sampling, unloading and other working personnel.	YES/NO	YES
28.	Whether there is an arrangement of control push buttons with necessary light indications for wagon coupling/decoupling staff at all WT/ SAC locations for generating a permissive for operation of the SAC.	YES/NO	YES
29.	Whether fire extinguishers are provided in Control rooms and staff is trained by CISF for handling/usage of same	YES/NO	YES
30.	Whether MGR/FT area is declared safe or not by Zone leaders	YES/NO	NO

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31.	Whether 360-degree surveillance CCTV Cameras with auto rotation feature should be installed at suitable locations in the unloading area and Exchange yard of MGR to have coverage / monitoring of maximum activities. Functioning of all CCTVs to be ensured all the time. (As per CC IOM ref. no. CCFT/MGR/Safety/01 dtd. 2nd July'20)	YES/NO	YES Available in TH and WT areas
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Responsibility: SCE (TH) & DGM (FT-O)

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13.3 Annexure – III

System of Train Working

Some of the common terminology used for train operation is given here -

- 13.3.1 Absolute Block Working (ABW)** – it is a system in which only one train/loco/Trolley/RRV is permitted to occupy the track at a time in the block. Block is the track between the Advance Starter (Last Stop Signal (LSS)) and the Home Signal (First Stop Signal (FSS)) of next station plus overlap. This system is implemented with the help of signalling system equipment (Axle counter, Block instruments etc.). In case of failure/non availability of signalling equipment, absolute block working can be implemented by following the private number exchange system and Authority to Proceed.
- 13.3.2 Private no. exchange system** – Private no. is a random two digit no printed in private no book with other details. ASM who want to dispatch train/loco, will call SM/ASM of receiving station for line clear. ASM of receiving station will give line clear only after ensuring the previous train/loco through the line/track sought by the dispatching station is completely received and has cleared the overlap of the home signal and he is ready to handle the next train/loco. Both ASM will exchange their private no and will record the others private no against their private no along with other details as per format. After receiving the line clear from receiving station, ASM of dispatching station will issue authority to proceed to Loco Driver, send the train/loco through the line agreed and will inform the ASM of receiving station about the departure. On complete receiving of train/loco ASM of receiving station will inform ASM of dispatching station about the arrival of the train. In case of non-availability of signals, station may proceed with private no. exchange with Loco Driver in place of authority to proceed only after due approval of same in LMI.
- 13.3.3 Authority to Proceed** – it is authority given to Loco Driver by SM/ASM for the movement of train in a block section with details like which line, which signals are not available and up to where the authority for movement has been given.
- 13.3.4 One signal one movement** – Any permission given (either by signals/ authority to proceed/ private no. etc.) shall be valid for movement of one train or other rail vehicle movement only. If another train / vehicle is to be moved in the section, then it shall be treated as separate train/vehicle and new permission needs to be taken from the station in advance after completion of first movement.
- 13.3.5 Station Limits** – It is the track section between First Stop Signal (Home Signal) and Last Stop Signal (Advance Starter) of the station. Within this limits, SM/ASM of the station has full authority for movement of train i.e., line clear from other stations is not required.
- 13.3.6 Block Section** – it is the track section between Last Stop Signal of first station

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and First Stop Signal and overlap of Next Station in the direction of movement. For movement in this section, line clear from station in advance is required by the station before sending a train.

13.3.7 Signals - A "Signal" is a medium to convey a particular instruction in nonverbal form to Loco crew. It can be a visual like fixed signals (colour light signals), (hand signals) and flare signals or audible like detonators, whistle, voice.

13.3.8 Aspect & Indication of Signals – -physical appearance of signal as seen by driver of train is called aspect. Indication is meaning of the aspect.

Sr No	Color of Signal	Aspect	Indication
01	Red	STOP	Dead Stop
02	Yellow/Amber	CAUTION	Proceed & be prepared to stop at next signal
03	Double Yellow/Amber	ATTENTION	Proceed & be prepared to pass next signal at restricted speed.
04	Green	PROCEED	Proceed.

Responsibility: SCE (TH) & DGM (FT-O)

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13.4 Annexure – IV:

IOM “MGR/FT system visit by higher officials.”



CORPORATE FUEL MANAGEMENT & TRANSPORTATION

INTER OFFICE MEMO

From: Executive Director(FM&FT)	To: HoPs of all Coal Stations of NTPC&JVs
Ref: CC/FT/2020	Copy: 1. REDs, ED(OS) 2. STA to D(O) 3. STA to CMD
Date: 20 th June 2020	

Subject: Inspection of MGR/Fuel Transportation System as per ORT Format

With a view to improve the overall safety and efficiency of MGR/Rail operations, in line with ORT format, stations have to follow the minimum target frequency for inspection of FT/MGR systems as prescribed in detail 6 of ORT formats and record the observations. The Knowledge Team members are verifying the record during Technical and Safety Audits.

Inspection Level	Frequency of Inspection(Minimum)
Head of MGR/ FT	Twice in a week
Head of Fuel Mgmt.	Once in Week
Head of O&M	Twice in a month
Head of Safety	Once in a month

*-50% inspections during night hours(23:00 hrs to 4:00 hrs)

It has been observed that inspection schedule for MGR/Coal transportation system is not being followed strictly in some stations. Seeing the importance of the inspections to ensure reliability and safety of coal transportation systems, it is requested to adhere to the inspection schedule prescribed in the ORT format, and observations during the inspections should be recorded and attended on top priority. The same should be reviewed for compliance in DPM/HOP reviews/ORT/MCM and with surprise inspections.

(M Prasanna Kumar)

I C O M I T

Responsibility: AGM (FT)

TITLE	Guidelines for MGR/Railway Siding Operation
LMI No.	LMI/OGN/OPS/SYST/035 Rev 03

13.5 Annexure – V:

IOM “FT circular for accident/derailment reporting”



CORPORATE OPERATION SERVICES

INTER OFFICE MEMORANDUM

From: ED(OS)	To: All HOP's (Coal Stations)
Copy (For information)	1. All REDs 2. STA to Dir (O)
Date	14 December 2022
Ref No.	COS/IOM/ED/22/063
Sub: Reg. reporting, restoration, and enquiry of accidents/derailments in MGR/FT systems by stations	

It is observed that all accidents/derailments in the MGR/FT systems are not being reported to OS-FT group by the stations. The restoration activities without due importance to investigation of attributable reasons obstruct in the system improvement and may results in repetition of similar accidents in other stations. Hence in-line with earlier issued CC-FT system circular - 01 / 2012 following may please be ensured by all stations on priority:

- Head of stations should form standing preliminary investigation team comprising of members from MGR-Operation, Rolling stock, Track, and Safety at site.
- Any accidents/derailments related to the MGR/FT system is to be reported to OS-FT group at CC-EOC Raipur too.

This is for kind information and necessary action please.

Ashwini Kumar Tripathy

Enclosed:

Annexure-I: FT system circular -01/2012 dtd 31.07.12

TITLE	Guidelines for MGR/Railway Siding Operation
LMI No.	LMI/OGN/OPS/SYST/035 Rev 03

NTPC Ltd.

(Corporate Fuel Transportation)

Date: 31st July 2012

FT- System Circular – 01/2012

Sub: MGR Accident –derailment Reporting, Restoration & Enquiry System

The system of reporting of accidents/derailments had taken a back seat at some of NTPC stations and the FM & MGR teams focus only restoration work without due importance to investigation of reasons which refrains the systems from improvement to safeguard against such incidents.

The guidelines for incidental reporting & Enquiry Committee constitution, enquiry recommendations & their implementation, in case of MGR related accidents / untoward incidents are framed as under:

A. Reporting of Incident:

whats app

1.0 Flash report of incident (through /email/sms) should consist:

- Location and sequence of event
- Rolling stock involved
- Injury if any
- Likely reason
- Approx. Loss
- Anticipated time for restoration – if applicable

2.0 Report to be flashed to RED, ED(OS), ED(FS),ED(FT) & D(O) within 01 hr. of incident.

3.0 Type of Incident:

3.1 Derailment with obvious reasons such as:

- a) Riding over the coal at Track Hopper / WT or at Track
- b) Wagon door opening
- c) Derailment which did not involve loss of rolling stock

3.2 Derailment/accident involving loss of rolling stock / MGR assets / Human life / run over.

4.0 Detailed records of all derailments should be maintained by Site MGR in a Derailment Register, which will be subjected to inspection during Tech. Audits.

— *S. Prakash*

Responsibility: AGM (FT)

TITLE	Guidelines for MGR/Railway Siding Operation
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B. Start of Restoration work

1. Head of Station should form a standing Preliminary Investigation Team comprising of members from Rolling Stock, Track, MGR-Operation and Safety. This team shall be in place and should reach to accident site as soon as possible and shall take up the activities as under:
 - 1.1 For incidents mentioned in A-3.1, the restoration work shall start immediately after investigation by any two members of Preliminary Investigation Team.
 - 1.2 For incidents mentioned in A-3.2, the restoration work should be started only after the clearance subsequent to critical inspection and investigation by Preliminary Investigation Team.
2. Preparation of site sketch depicting the direction of movement, position of derailed/capsized rolling stock, identifying & displaying the point of drop/mount, displaying the markings on sleepers track etc. & assessment of the probable primary reasons of incident should be done by Preliminary Investigation Team.
 - 2.1 Recording of track parameters of 30 meter track on both sides of the accident site, Measurement of Wagon & Loco wheel profiles of the involved rolling stock wherever possible, their coupler conditions, S & T system conditions near site and recording of brief statements of the driver/guard/station staff/points-man etc.
3. After assessment, the Investigating Team shall allow the site staff/Station FM to start restoration work and shall prepare a report which shall be submitted to the Enquiry Committee for subsequent investigations.
4. Site FM should Plan & mobilize for restoring of coal movement in such an event which may involve single line operation / movement in reverse direction / slow speed cautioned movement etc. keeping safety practices in mind
5. Site FM should restore the system to normalized state & Traffic operations to be restored to normal operations.
6. Site should Inform the system normalization to corp. OS, Corp. FS & FT & regional Office.



TITLE	Guidelines for MGR/Railway Siding Operation
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C. Constitution of Enquiry Committee & Reports on Enquiry

- 1) In line with Corp. Operation directive (COS-ISO-00-OD/GEN/004 Rev. No.:2, October, 2005); the Enquiry Committee should be constituted by Competent Authority (C.A.) within 24 hrs of reporting of incident.

The responsibility of formation of enquiry committee is as under:

- a. Corporate FT Deptt. : where C.A. is D(O)
- b. Regional OS/FM : where C.A. is RED
- c. Station MGR/FM : where C.A. is HOS.

- 2) Enquiry Committee should consist at least two members (from MGR stations/RHQ/CC) having sufficient working experience and knowledge of MGR & Railway systems in order to have qualitative investigation of the incidents.

- 2.1 Enquiry Committee should inspect the site & start investigation as early as possible. The collected data and report of Preliminary Investigation team should be given to Enquiry committee.

- 3) Enquiry Committee should investigate/enquire on various technical, operation & safety aspects of incident and thereafter should submit the report to C.A. in the following format:

- 1) Pre-ample
- 2) Committee Constitution details
- 3) Time schedule
- 4) Enquiry Proceedings
 - 4.1) Description of Accident site (with photo, sketch etc.)
 - 4.2) Details of track, S&T etc.
 - 4.3) Details of documents & records checked.
 - 4.4) Investigation & committee observations.
 - 4.5) Establishment of reason of accident
 - 4.6) Cost of damages.
- 5) Recommendations / Remedial measures suggested.

- 4) In case, the accidental incidents are repeated, at any station, an external independent agency should also be engaged in the investigation



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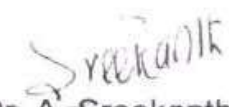
D. Submission of Report & follow-up

1. Report to be submitted to the competent authority by the leader / Head of Enquiry committee
2. Regional OS/FM/ORT coordinator should monitor the implementation of enquiry recommendation after approval, through various meetings such as monthly ORT.
3. Regional ORT Coordinator shall forward the Enquiry Report to CC-FT, for circulation of *Recommendations of the Enquiry Committee* to other stations / uploading the same on Web page of Corporate-FM / MGR, for avoidance of such incidents by modifying the practices by other stations as per enquiry findings.
- 3.1 The Enquiry Closure Report to be submitted by Regional OS /FM, to CC-FT stating the reasons & responsibilities as per the findings of the Enquiry report.
4. Station FM shall maintain and keep the records of all enquiries and the enquiry reports after implementation of recommendations and shall be the custodian of the same.

E. Implementation

1. Head of Station (HOS) shall be responsible for implementation of Enquiry Committee report recommendations.

This issues with the approval of Competent Authority.


 (Dr. A. Sreekanth)
 Executive Director (FT)

Distribution :

1. All HOS-Coal Stations
2. All REDs
3. ED (OS)/(FS)
4. D (O): For Kind inf.

Responsibility: DGM (FT-O) & AGM(FT)

TITLE	Guidelines for MGR/Railway Siding Operation
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Derailment/Incident Flash report Format

(to be reported through Message/WhatsApp/Email within 1 Hr of incident)

Ref: -

Date.....

Respected Sir,

(D(O), RED, ED(OS), ED(FM), ED(FT)),

An incident/accident happened in NTPC at HH:MM Hrs, on
XX.XX.20XX

Incident/Accident brief is as following for kind information:

S. No	Description	Particulars / Remark
01	Loco No (IR/NTPC):	
02	Wagon No (IR/NTPC):	
03	Position of Wagon from Loco (sl. no.):	
04	Location of derailment:	
05	Injury/Fatal, if any	
06	Loss of assets (approx.)	
07	Restoration work / completion expected by HH:MM Hrs Dtd DD.MM.YY.	
08	Likely prima facie reason:	

Thanks & Regards,

(Shift I/c / MGR I/c)

Responsibility: DGM (FT-O) & AGM(FT)

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Derailment/Incident Detailed Enquiry Report Format

Ref: -

Date.....

The detail of investigation is as follows –

1. Preamble –
2. Committee Constitution Details –
3. Time Schedule –
4. Enquiry Proceedings –
 - a. Description of Accident Site (with photo, sketch etc.) –
 - b. Details of Track, S&T etc. –
 - c. Details of documents & records checked –
 - d. Investigation details & committee observation –
 - e. Establishment of reason of accident –
 - f. Cost of damages –
5. Recommendation / Remedial measures suggested –

Committee Members –

1.

2.

3.

4.

Responsibility: DGM (FT-O) & AGM(FT)

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Derailment/Incident Detailed Enquiry Report Format

Ref: -

Date.....

The detail of investigation is as follows –

1. Preamble –
2. Committee Constitution Details –
3. Time Schedule –
4. Enquiry Proceedings –
 - a. Description of Accident Site (with photo, sketch etc.) –
 - b. Details of Track, S&T etc. –
 - c. Details of documents & records checked –
 - d. Investigation details & committee observation –
 - e. Establishment of reason of accident –
 - f. Cost of damages –
5. Recommendation / Remedials measures suggested –

Committee Members –

1. 2. 3. 4.

Responsibility: DGM (FT-O) & AGM(FT)

TITLE	Guidelines for MGR/Railway Siding Operation
LMI No.	LMI/OGN/OPS/SYST/035 Rev 03

13.6 Annexure – VI: “Compliance of recommendations for safety and reliability in FT/MGR



Corporate Fuel Management & Transportation

INTER OFFICE MEMO

From: Executive Director (FM & FT) edft@ntpc.co.in , edfm@ntpc.co.in	To: HOPs (All coal based stations including JVs/Subsidiaries)
Ref. No.: CCFT/MGR/Safety/01	Copy for kind information please: ED(OS)/ED(Safety) REDs STA to D(O) STA to CMD
Date: 2nd July, 2020	

Sub: Compliance of recommendations for safety and reliability in FT/MGR area



This is in continuation to earlier communications regarding system compliance in Fuel Transportation and Fuel Management areas (Copies attached). Safety which is one of our core values is crucial part of our work life. We should accord top priority to safe working environment and strive hard for zero incidences/accidents at work.

Stations should take proactive actions for safe working, learn from the experiences and incidences that happened in NTPC/other stations and sustain their compliance with regular monitoring. Stations should also adhere to safety guidelines, procedures and standards in Fuel Transportation/MGR area, and ensure that such incidences don't repeat.

A gist of general recommendations from previous incidences/derailments in MGR area is attached at Annexure-A, for necessary action and compliance at the earliest. The compilation is related to safety, operation philosophy, reliability and systems development and adherence which need regular review to achieve accident free MGR operations in NTPC. The compliance needs to be ensured immediately and the same should be reviewed at different levels.

The compliance status of these recommendations may be communicated to CC-FT by 10th July 2020 and subsequently by 10th of every month for review and generating MIS with deviations / exceptions.

Encl: As Above.

(M.PRASANNA KUMAR)



area”

TITLE	Guidelines for MGR/Railway Siding Operation
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**General Recommendations from previous
accidents/derailments in MGR for Immediate Compliance**

Description of the Issue	Present Status-Compliance	Action taken if not complied	Target date
TRAINING/MANPOWER			
1. Development/Refresher Training should be provided to the operating staffs which include Loco Pilots, ALPs, Station controllers and points man/shunt man etc, so that they should well aware of safety requirements and safety consciousness. The refresher courses can be organized in in-house training institutes attached to the power station by the MGR senior executives /the safety officer of the project / Railways Divisional Safety Officer at regular interval.			
2. New recruits by contractor should be tested/interviewed thoroughly and for develop adequate skills before assigning independent responsibility/Duty.			
3. The engagement of retired railway staff (drivers, controllers and other cadre) should be strictly not beyond 65 years of age in any case.			
OPERATION			
1. Running staff (Loco Pilots/ALPs) working hours should not exceed 10 hrs at a stretch keeping in line with the safe Railway working procedure.			
2. Prescribed medical examination of the Loco Pilots and other staffs should be regularly carried out in time.			
3. An electronic breath analyser should be made available in the MGR- and necessary monitoring should be carried out regularly (to Loco Pilots, ALPs, Station controllers and points man/shunt man etc for each and every duty. For other contract workers on random basis). The breath analyser results should be recorded in a register and should be counter-signed by the NTPC-Shift in-charge. This is to ensure grade reported for duty are not under the influence of alcohol (at the time of reporting for duty or during their duty period.) If any result found beyond limits the person should not be given duty on that day and if the same is repeated with the same person for 3 times he may be terminated from the duties permanently.			
4. No shunting to be done without point man (shunt man) and without proper wireless/Mobile communication.			
5. The Loco Pilots/ALPs should obey speed restrictions/caution orders. Necessary surprise check needs to be exercised to insure the same.			
6. No rake should enter the block section until and unless ongoing rake has cleared the block section and cleared the safe distance beyond the 1 st stop signal of the next station.			
7. Pep talks, awareness safety talks and safety pledges may be carried out on daily basis. Periodic medical health check-up for workers involved in rake movement operation			

TITLE	Guidelines for MGR/Railway Siding Operation
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**General Recommendations from previous
accidents/derailments in MGR for Immediate Compliance**

Description of the Issue	Present Status- Compliance	Action taken if not complied	Target date
may be ensured. Record of health check-ups to be maintained suitably.			
8. In every shift, a supervisor or NTPC Operator/Technician shall be available in WT area to monitor the activities of unloading throughout the shift and maintain a necessary Logbook.			
9. Shift rota of Contract workmen deployed in MGR operation for tipping assistance shall match with CHP Operation shift rota for a better coordination between the on duty MGR & CHP groups.			
10. Control push buttons with necessary light indications to be provided at all SAC locations of the operators and apart from manual signalling these push buttons are to be operated for generating a permissive for operation of the SAC. Provision of Human sensor on/for SAC movement is to be expedited.			
11. Tipping shall be avoided if deployment of sufficient manpower is not met at inhaul/overhaul side, SAC, WT control room etc.			
TRACK			
1. Coal accumulation along the tracks & other foreign/ loose materials if any are to be removed on regular basis. Proper Cess to be maintained along the sides of the track for ease of movement at required.			
2. USFD testing of rails and Weld Joints for the entire length of the track is to be carried out at an interval of 6 months. Proper record to be maintained. Defects observed during USFD testing of rails/Welds to be attended if the defect reports IMR.			
3. Creep adjustment is to be carried out yearly or as and when required and avoid cut rails less than 6.5 mtr/ defective rails/wheel burns on the track section			
4. Brick or Concrete Pathway on both sides of the track along with proper distributed lighting for the entire length of the Rake to be provided for safe movement of pointsmen/Shuntmen on both inhaul side and outhaul side. The areas of operator movement at SAC inhaul and outhaul side are to be properly brick paved or concreted inside the both rails of the track for a minimum length of two wagons.			
5. Track fitness certificate for rolling should be given by competent authority (track engineer in consultation with PWI/AEN) and the speed restriction advised by the competent authority should be strictly adhered.			
6. All the times Points and crossings to be maintained Healthy through proper regular and special maintenance (Replacement/ Reconditioning of Tongue rails)			

TITLE	Guidelines for MGR/Railway Siding Operation
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**General Recommendations from previous
accidents/derailments in MGR for Immediate Compliance**

Description of the Issue	Present Status- Compliance	Action taken if not complied	Target date
SYSTEM COMPLIANCE			
1. The LMI/SOP to be made available for MGR as well as WT+SAC operations and needs to be revised as and when felt th scope for improvement to cover all the aspects of operations and signalling including manual interventions.			
2. SOP of POH/IOH of wagon/Loco to be followed strictly as per Railway norms.			
3. A checklist of safety/ protection and standard operating Instructions including Do's/Don't should be prepared in Hindi/English and Most importantly in local language (is Must) to be displayed in MGR and CHP areas. The same should be reviewed time to time. The Clear visibility of these display at all times must be ensured.			
GENERAL & SAFETY ISSUES			
1. 360 degree surveillance CCTV Cameras with auto rotation feature should be installed at suitable locations in the unloading area and Exchange yard of MGR to have coverage/monitoring of maximum activities. Functioning of all CCTVs to be ensured all the times.			
2. The Wooden Wedges used to prevent the wagon roll back, should have a longer and comfortable handle so that the wedge can be handled from a safe distance from the rake/rail track			
3. Possibility of installing photo sensor or wave sensor or any other means may be explored to sense human presence or movement across the track at WT area if any to immediately stop the SAC movement.			
4. Installation/Display of more caution signs against electric hazard near OHE line in exchange yard to be installed in order to improve awareness among the MGR working personnel.			
5. Any unmanned railway level crossings in MGR area to be manned and to be provided with Audio visual alert system to the road traffic along with speed breakers , Boom barriers and height gauges as per standard railway safety system.			
6. Illumination in Wagon Tippler/Track Hopper area should be adequate.			
7. Whenever wagon or SAC coupling knuckle is to be adjusted, it shall be done with a suitable "L- lever" of sufficient length from outside the track without entering into the Rails.			
8. The side arm charger inhaul side coupling maintenance practice needs to be reviewed, coupling healthiness to be seen after each rake unloading/tipping is complete.			
9. Prior approval of competent authority is to be obtained before any modification or new provision in existing system of MGR with sketch and operating instructions.			

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**General Recommendations from previous
accidents/derailments in MGR for Immediate Compliance**

Description of the Issue	Present Status- Compliance	Action taken if not complied	Target date
10. Foot Over Bridge/RUB to be constructed at a suitable location wherever required for the safe movement of the work force.			
11. At Track Hoppers Supervision to be improved by Agency to ensure the Unloading workers are working at the safe distance from the rake and a provision of hooter warning to be provided for any rake movement in each placement before giving clearance to the loco driver.			
12. There should be HIRA (Hazard Identification, Risk Assessment and Control Measures) and JSA (Job Safety Analysis) in place. The safety checklist shall be enforced for implementation.			
13. Key areas prone to unauthorized stoppage of train by villagers may be identified and attended by cordoning/strengthening security arrangements.			
14. Inspections including footplates and monitoring of installations/equipment, staff should be carried out and must be programmed as per ORT format. At least 50% foot plate inspections should be carried out during night time (23:00 hrs to 04:00 hrs).			
15. Senior Executives (MGR-I/c & FM-I/c) should frequently inspect level crossings, rake stopping places, examine the staffs with reference to working instructions etc.			
16. Prescribed Track inspection schedules by the plant top management shall be adhered.			

TITLE	Guidelines for MGR/Railway Siding Operation
LMI No.	LMI/OGN/OPS/SYST/035 Rev 03

13.7 Annexure VII:

IOM "Supervision of Rail Operation by Heads of MGR/FT"



Corporate Centre

Corporate Fuel Transportation

Inter Office Memo

From: ED(FT)	To: Heads of MGR/FT
Ref: CC/FT/2020/01	Copy to: 1. HOPs of Coal Stations 2. STA to D(O) – for info of D(O)
Date: 19.03.2020	
Subject: Supervision of Rail Operations by Heads of MGR/FT	

With a view to make MGR/FT operations safer and efficient, it is advised that all Heads of MGR/FT shall visit MGR Control Room on day to day basis (preferably in mornings) and spend adequate time to monitor the O&M activities of the day.

They are advised to take stock of the operating conditions including –

- Operations as per LMI/SOPs,
- permits issued,
- caution orders for the day,
- S&T availability and signal/points bypassing,
- track activities,
- rolling stock status including VCD, speed recorder, last vehicle brake pressure etc.,
- PM status of rolling stock, track, points and crossings etc.,
- manpower deployment (including in en-route stations),
- working hours of running staff (including over-time status),
- cycle time of previous day rakes including speed monitoring,
- loading/unloading delays,
- demurrage status,
- rake pipeline (in case of IR rakes)
- interact with operational staff and record their observations in the shift logbook

In addition to this, they are also advised to conduct periodic in-situ inspection of all maintenance and contract related activities and infrastructure including workshop, intermediate stations, track, rolling stock, S&T (relay panels, batteries) and relief van.


(N Srinivas)
IRTS

TITLE	Guidelines for MGR/Railway Siding Operation
LMI No.	LMI/OGN/OPS/SYST/035 Rev 03

13.8 Annexure VIII:

Do's & Don'ts for VHF equipment

How to Use VHF Equipment

Dos and Don'ts

VHF equipment (Walkie-Talkie & Base Station) are one way radio i.e. at a time only one person can Talk and all other have to Listen. It is just like single line operation of Rail Track where at a time only one Train is allowed to occupy the track.

Dos –

1. Ensure that your VHF equipment is tuned to **proper Channel**.
2. Ensure that **channel is free** before pressing the PTT button (starting communication) i.e. others are not communicating. If someone is communicating please wait till their communication is completed.
3. Ensure that PTT button is **pressed and hold** before start speaking and keep it pressed while speaking. Release it only after you have stopped speaking.
4. **ALWAYS** speak the name of person/Control Station first to whom you want to communicate with. Also if you have received some message/communication **ALWAYS** acknowledge it.
5. Keep your message (communication) **short** i.e. break your message in short sentences in place of one long sentence.
6. **Wait** for some time before repeating message again, so that other can respond. Give a little pause between two communications.
7. When shutting down Loco, **switch off** Base Station too.
8. Give priority to **placement activity** at Track Hopper. If it is in progress, wait as far as possible, to completed it.
9. Ensure that Walkie-Talkie is sufficiently charged before issuing it. Enter it in register provided in Control Room and before leaving from duty return it and cancel it from register.

Don'ts

1. Do not press PTT button while someone is communicating, as it will disturb communication in process, even if you have not spoken.
2. Do not keep more than one equipment ON around you. It will echo (ring) your voice while you are talking on VHF.
3. Do not respond in hurry even if you have got the message. Let the first person complete his communication, pause a bit and then respond.
4. Do not hold/bend/twist the Antenna of Walkie-Talkie, it will damage it.
5. Do not remove the dust cover of programming port of Walkie-Talkie.



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वीएचएफ उपकरण की उपयोग विधि

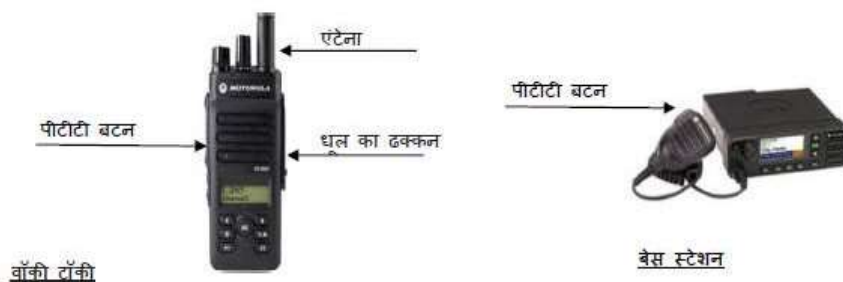
क्या करें और क्या नहीं

क्या करें :-

1. सबसे पहले यह सुनिश्चित करें कि VHF उपकरण उचित चैनल (Channel) में लगा है।
2. बात शुरू करने से पहले यह सुनिश्चित करें कि VHF में कोई अन्य बात नहीं कर रहा है। यदि कोई बात कर रहा है तो कृपया उनके बात पूर्ण होने का इंतजार करें।
3. बोलना शुरू करने से पहले पीटीटी (PTT) बटन को दबा कर रखें और उसके उपरांत ही बोलना शुरू करें। बोलना बंद करने के उपरांत ही पीटीटी बटन को छोड़ें।
4. जिनसे बात करना है उनका नाम/कंट्रोल स्टेशन का नाम संबोधित करते हुये ही बात करें। साथ ही यदि आपको कोई संदेश/आदेश प्राप्त हुआ हो तो इसकी प्रतिउत्तर अवश्य दें ताकि सामने वाला सुनिश्चित हो।
5. कोशिश करें कि आपका संदेश संक्षिप्त हो (लंबा वाक्य के बजाय छोटे छोटे वाक्यों में बोलें)।
6. अपना संदेश दोहराने से पहले थोड़ी देर इंतजार करें ताकि सामने वाला व्यक्ति को जवाब देने का पर्याप्त समय मिले। दो वाक्य के बीच में थोड़ा सा अंतराल रखें।
7. लोको बंद करते समय बेस स्टेशन (Base Station) को भी बंद करें।
8. यदि ट्रेक हॉपर (Track Hopper) में प्लेसमेंट (placement) कार्य जारी हो तो उसे प्राथमिकता दें, और यह कार्य पूर्ण होने तक इंतजार करें।
9. वॉकी टॉकी लेने से पहले सुनिश्चित करें कि यह पूर्णतः चार्ज है। इसकी सूचना कंट्रोल रूम में उपलब्ध पंजिका में अवश्य दर्ज कराएँ और कार्य मुक्त होने पर वापस कर पंजिका से अपना नाम हटवायें।

क्या नहीं करें :-

1. यदि कोई अन्य बात कर रहे हैं तो पीटीटी (PTT) बटन न दबाएँ, ऐसा करने से उनका संवाद प्रक्रिया बाधित होगी, चाहे आपने कुछ बोला हो या नहीं।
2. आपने आस-पास एक से अधिक वीएचएफ उपकरण चालू न रखें। ऐसा करने से आपकी आवाज गुंजेगी।
3. जल्दबाजी में सामने वाले के बात पूर्ण होने से पहले पीटीटी बटन न दबाएँ भले ही आपको संवाद समझ में आ गया हो। सामने वाले को बात पूर्ण करने के उपरांत ही बात करें।
4. वॉकी टॉकी के एंटेना को पकड़ कर न रखें और न ही इसको मोड़ें अथवा मरोड़ें। ऐसा करने से यह खराब हो सकता है।
5. वॉकी टॉकी के प्रोग्रामिंग द्वार (programming port) के धूल का ढक्कन को न हटायें।



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13.9 Annexure IX:

Monsoon & Summer Preparation

MGR/FT area Monsoon preparation & precautions (1st May to 15th June)

S.N.	DETAILS OF PREPARATION	ACTION PROPOSED
A. Operation		
1	Strengthening of unloading manpower at track hopper.	Ensure availability of additional manpower, if required, for timely unloading of wagons with sticky coal.
2	Draining of moisture from Loco air reservoir.	After onset of Monsoon, ensure draining of moisture from Loco air reservoir by Astt. Loco drivers during takeover/handover, where the automatic drain valves are removed, and cocks are provided
3	Regularly filling of sand in Loco.	Ensure that sand is regularly recouped in sand boxes of all the Locomotive during "Safe to run" testing before every trip. Ensure entry in Driver's log book regarding functioning of sanders and availability of sand. A record of instances of stalling of Loco due to wheel slip should be maintained for planning of corrective actions.
4	Ensure Umbrellas, Raincoats availability in all control rooms.	Procure new Umbrellas and Raincoats, if required.

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MGR/FT area Pre-Summer preparation & precautions (15th March to 15th June)

S.N.	AREA	DETAILS OF PREPARATION	ACTION PROPOSED
1.	General	General preparedness	Clear and bold display of contact number of fire station at different locations.
2.			Wide communication of summer related information like Heat stroke, Sun burn etc. and their prevention through pep talks with Contract workers.
3.			Awareness of Emergency response/ Disaster management Plan (especially to the control room executives) to handle any unforeseen emergency
4.		Firefighting System	Schedule Fire mock drills with CISF and arrange training for using fire extinguisher with all workers of all sections including Loco Drivers & Control room staff of all shifts.
5.			Ensure availability and healthiness of portable fire extinguishers as per deployment plan.
6.			Necessary arrangements should be in place to detach the wagon and quench the fire at isolated place, in case, coal in any wagon catches fire.
7.	Operation	Sensitization of Operating staff for fire hazards	Drivers/ operating staffs must be alerted to report any signs of abnormality in track and availability of dry grass/ inflammable material in adjoining area of MGR/FT circuit.
8.			Drivers must be alerted to check the Loco thoroughly, before taking over, for any leakages of lube oil/ HSD in various systems.
9.			Drivers must be alerted to feel the axle boxes as often as possible and their temperature should be monitored at various out pits / sheds on return.
10.			Ensure availability of proper type of fire extinguisher in all Locomotives & control rooms

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13.10 Annexure X:**ANNEXURE X****DISASTER MANAGEMENT PLAN****1. TERMINOLOGY****ACCIDENT:**

Accident is an occurrence in the course of working/functioning of MGR, which does or may affect the safety of MGR, its permanent way, works, engine, rolling stock, persons or which may affect the safety of others (whether connected or not connected with MGR) or may cause delays to trains.

SERIOUS ACCIDENT:

Serious accident is an accident which occurs in the course of working of MGR and is accompanied with the loss of human or grievous injury to the person or serious damage to the property or serious interruption to traffic.

TRAIN ACCIDENT:

Train accident involving a train which may be a set of loaded or empty vehicles worked by a Locomotive when running under a particular number or/ and name from a point of departure to a destination.

INJURIES:

Injuries are of three: 1. Grievous, 2. Simple and 3. Trivial.

SERIOUS DISLOCATION OF TRAFFIC:

Traffic is said to be seriously dislocated if main line is blocked for movement for more than 2 hours.

SABOTAGE:

Sabotage means the criminal interference with any part of the working machinery of MGR with the intention to damage the MGR property other than train wrecking or attempted train wrecking.

TRAIN WRECKING:

This means the willful tampering or obstruction with the permanent way or works, bridges or rolling stock, resulting in accident to a train with or without loss of life or damage.

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COLLISION:

When a train collides head-on or from rear against any train or vehicle whether moving or stationary is called COLLISION. Collision includes a side collision. But train collision with road vehicles at level crossing shall be termed as level crossing accident.

2. GENERAL STANDING ORDERS FOR DISASTER MANAGEMENT OF MGR-SIMHADRI**DISASTER MANAGEMENT:**

Disaster management means handling/managing the unsafe/unwanted situations in planned, safe and systematic manner with minimum time in case of emergency.

1. SERIOUS INJURY:

Loss of human life or part of the human body is injured or damaged.

2. LOSS OF MGR ROLLING STOCK:

This means damage of Railway track by derailment, damage to Locomotive and / or wagons etc.

3. FIRE:

Occurrence of severe fire to MGR rolling stock or within the premises of MGR area i.e. damage to Loco, wagon (empty or loaded), etc.

4. FLOOD:

Wash out of rail track embankment partly or completely where movement of rake is totally unsafe.

Disaster management may be tackled in the following manner as and when required.

1.1 GENERAL INSTRUCTIONS

Whenever any MGR person receives any information regarding an accident by any means, he shall immediately inform to the MGR Control Room shift charge engineer and all possible assistance to be rendered to the casualties and try to save the life of injured person if any and MGR system.

1.1.1 GENERAL DUTIES OF SHIFT CHARGE ENGINEER(TH) IN CASE OF AN ACCIDENT

When the shift charge engineer gets the information of an accident by any means he shall ensure the following:

1. In case of derailment at track hopper, shift charge engineer shall ensure that level of coal bed at the affected area is maintained at the appropriate level to facilitate safe re-railing operation.

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2. Arrange for immediate dispatch of the nearest medical assistance, if required.
3. Arrange for any other assistance as may be required at the site of accident.
4. Report the accident to all concerned as required, with time of accident and other detailed information.
5. Start mobilizing activities for the relief van.

1.1.2 GENERAL DUTIES OF MGR OPERATION IN-CHARGE AND MGR IN-CHARGE OR SENIOR MOST OFFICER OF MGR

1. On receiving the information of an accident, he should contact TH Control Room (if message is received from other sources) and rush to the accident spot immediately. He shall take charge of the situation and try to guide shift charge engineer till injured persons are shifted to the hospital and other assistance arrives at the spot for clearing the track for normal movement.
2. After quick inspection of the site of the accident he shall ensure that all the possible medical help is provided to the injured person/s, arrange for their transportation to the nearest hospital.
3. In case of fatal accident due to run over, immediately pass on the information to HR department (if possible, name, village and other information regarding the deceased and location of run over etc.). Information is also to be given to the head of CISF and local police station in consultation with HR group.
4. Telephone nos. of police station in the vicinity are as the under:
Parawada Police Station --- 08924-247233
5. **Arrangements to be made for the following:**
 - a. Photographer to taking evidence proof.
 - b. Arrangement for stretcher, markeen cloth, sweeper etc.
 - c. Arrange for truck/ambulance for shifting dead body for post-mortem. Shifting of deceased is to be done in coordination with local police and under their instructions only.
 - d. Note the name of the deceased person, his father's name, age, address, relative's names etc. after the identification is done by the police.
 - e. Inform Insurance person through central stores.
 - f. If situation goes beyond control due to accident / unwanted incident, availability of CISF and police personnel is to be ensured to protect the property of MGR.
6. In case derailment, collision etc., make a quick survey of the spot, note the details of damage, type of accident, possible cause/s of accident, name of the staff involved during the course of accident. Check the track in the adjacent area of the derailment / **collision** and ensure the possibility for movement of relief van to the nearest position of the accident / collision.
7. All the staff of MGR is advised to refrain from press and general public and not to spread any rumours.

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8. MGR Operation in-charge to ensure the following at the earliest in consultation with TH Control Room:
- Block the track from both the ends by putting red banner flag at both ends where derailment has occurred.
 - Arrange the photographer to take photographs of the accident spot for future use for insurance and / or enquiry etc.
 - Keep the coal rake movement under control if the derailment is in one of the Track Hopper and other is available for operation. Guide the shift charge engineer in this regard and ensure safety of the rolling stock during emergency operation.
 - Isolate the affected wagon from both ends and make approach possible for the rerailing activity.
 - Rescue team along with the re-railing equipment to be shifted to the affected spot through un-affected portion of the track at the earliest.
 - Supervise the relief work and provide all assistance for transportation or movement of Loco on the same if required. Arrange first aid, tea, snacks, communication etc. as may be required for the persons involved in re-railing work.
 - After re-railing work is over, the track is to be inspected by the track maintenance group. Obtain fitness certificate from them for normal or restricted movement after they have inspected and attended to major / minor defects. Check the gauge.
 - Clear the track for normal movement of coal rake and restore the normal movement.
 - Inform all the Sr. Executives: AGM (FM), GM (FM) AND GM (O&M) REGARDING clearance of track.
 - Ensure that the re-railing equipment is brought back and placed in the DOC-House and safeguard its key.

1.2 In case of fire:

Below standard instructions are to be followed:

- If fire is observed in Loco or wagon in moving train, immediately inform to the driver and TH Control Room. The driver has to stop the train immediately.
- Isolate the Loco / wagon from rest of the rake (if possible)
- If possible, place the Loco or wagon near the source of water or accessible to fire tender.
- Rush to the site with fire tender.
- After extinguishing fire, keep the burnt coal loaded wagon at R&D yard separately and pour water further. Then unload the same in presence of shift charge engineer (TH) and CISF fire personnel.

1.3 RESPONSIBILITIES OF HEAD OF MGR:

- Co-ordinate all the activities of re-railing operation and ensure speedy and safe clearance of track for normal coal rake movement.
- Co-ordinate with other departments case of requirement during re-railing operation.
- Co-ordinate with state authorities or Indian Railways, if required.

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4. Keep informing the TH Control Room about the activity at site.

1.4 RESPONSIBILITIES OF HR DEPARTMENT:

1. Maintenance of IR in case of local mob is disturbing the re-railing operation.
2. Ensure law and order with the help of CISF and if required local police to safeguard the property of MGR system.
3. In case of fatal accident due to any reason, ensure speedy clearance of track in co-ordination with the local police.
4. Co-ordinate for other formalities in case of fatal accident like post-mortem and handing over the body to the deceased relatives.

1.5 RESPONSIBILITIES OF SAFETY DEPARTMENT:

1. To assist in safe rescue / relief / re-railing operation.
2. To arrange required safety equipment.

1.6 DUTIES OF MGR OPERATION INCHARGE:

2. Ensure block for the affected area till clearance is received from track maintenance or MGR in-charge for normal movement.
3. Whenever required, arrange to shift the relief material to site through MGR circuit where road approach is not feasible.
4. If required, seek help from other departments for the resource management.
5. Ensure sufficient illumination at the affected area.
6. Be careful in movement of the train as it may involve movement in the reverse direction also. All the Loco crew should be made aware of the situation and should know the plan of movement.
7. Continuously keep in touch with the Loco crew in circuit.

NOTE: No train movement or Light Loco movement to be permitted till clearance to such effect is received from the track maintenance in-charge in writing.

1.7 RESPONSIBILITIES OF TRACK MAINTENANCE INCHARGE:

2. Take block permit for the affected area for safety of the working persons and rolling stock.
3. He must be equipped with the wireless communication when at site.
4. Give requirements for relief operation to the TH Control Room or FT Operation in-Charge.
5. Identify the areas to attack on priority to ensure early revival of coal rake movement (may be in a skeleton manner in the beginning) and start the work on those lines.
6. Ensure normal movement of coal rake at the earliest possible.
7. Take all the safety precautions required for the work.

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3. REPORTING OF ACCIDENT:

1. On receiving the information of an accident, the station in-charge must immediately report the accident to the concerned persons.
2. The date, time, Loco no., wagon no., names of the Loco crew, nature of accident and all the relevant information should be recorded in the logbook.
3. Anticipated duration for which track will remain block and approximate total length of track damaged and / or blocked by accident should be reported to the concerned persons.
4. Information is to be given to the HOD of TS, safety, HR, C&M, Police Station, insurance etc. as the case may be.
5. The incident / accident message to the above persons should be brief but clear and should contain the type of assistance / help required.
6. In case of serious accident or in cases of accident to Light engines, lorries, trolleys or fire in diesel / electrical system resulting in death, grievous hurt or loss to the extent to be noted in the accident and to be informed to the concerned officials.

4. FLOOD OR LAND SLIDING:

Whenever any Loco crew or patrolling staff notices flooding of track or land sliding, he shall inform R&D/TH Control Room immediately. Patrolling person shall inform to the first passing train by stopping it using red flag or red Light in case of night hours. The Loco driver will stop the train at safe distance from the spot and inform to TH Control Room, shall walk down up to the spot, if possible, and give feed back to the Control Room regarding the case and the help required.

Shift charge engineer on receiving the information about flood or land sliding, shall ensure stoppage of movement of train in the affected area and shall inform to track maintenance in-charge and operation in-charge. He shall also ensure the blockage of the section and shall try to provide necessary help at site wherever possible.

5. PREPAREDNESS OF MGR:

1. At least 2000 to 3000 gunny bags must be kept ready filled with sand to start immediate relief work in case of land sliding or flood. Further requirement of filled gunny bags can be judged during the course of relief operation and can be arranged.
2. Around 100 to 150 numbers of bullies of about 6 meters height and 6" diameter should be kept readily available.
3. These items should be available at a suitable location within the workshop premises.

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6. IMPORTANT PHONES OF SIMHADRI

Sl. No.	Designation	Contact No.		
		Office		Cell/Mobile
		RAX	BSNL-Res	
1.	CGM OFFICE	4001/4002	243001	
2.	GM (O&M	4251	243960	
3.	AGM (FM) – I/C	4402	243431	8332988907
4.	AGM (FM)	4284		9425534368
5.	AGM (HR)	4074		9650991250
6.	DGM (FT)	4212		9442100369
7.	Crossing Station	4917		7702955539
8.	R&D Cabin	4395/4397		9440918546
9.	TH-C/R	4299/4399		9440918547
10.	CHP-C/R	4500/4600		7901252614
11.	CISF Fire Station	101/4009/6101	243573	8332988843
12.	CISF Commandant	4008		
13.	First Aid Center	4460/6102	243217	7901695390
14.	Hospital	102/3409		7901694390
15.	Safety Control Centre	4023/4025/6103		9491876071
		4962		

Parawada Police Station: 08924-247233

